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Modelling and Simulation of Vehicle-to-Grid for the Large-Scale Integration of Offshore Wind Power in New York City

Master Thesis

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Abstract

This thesis explores the potential of vehicle-to-grid (V2G) technology in the context of New York City. Specifically, the work aims to model mobility behavior and offshore wind power to assess the feasibility of energy storage enabled by V2G. Firstly, I outline the methodology for simulating mobility patterns and the power output of wind farms. Subsequently, I outline a preliminary strategy to synchronize mobility, wind power availability, and the charging and discharging activities of electric vehicles (EVs). The thesis analyses several scenarios and identifies system limitations to ensure realistic simulations within computational constraints. These findings form the foundation for an innovative smart-charging algorithm, which examines the underlying dynamics governing EV fleets participating in V2G. Finally, I present a refined smart−charging algorithm calculating the actual number of EVs participating in V2G when integrating offshore wind power sources. The primary objective of this research is to gain a broad understanding of V2Gs true technical potential and comprehend the dynamics that govern this system. In conclusion, the study contributes to the sustainable development of coupled mobility and energy systems by providing insights into the potential of V2G technology to facilitate energy storage and reduce carbon emissions.

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Chapter 1

Introduction

The world faces a severe climate crisis, which calls for a thorough reconsideration of our energy systems and a determined pursuit of sustainable alternatives. We need to significantly transform our energy production and consumption habits to reduce global warming in the next three decades. This transformation involves decreasing our reliance on fossil fuels, increasing the use of low and zero-carbon energy sources, and exploring alternative energy options. One important aspect of shifting towards sustainable energy solutions is closely linked to our cities. Urban areas are crucial in the transition towards sustainable energy solutions because they are the epicenters of human activity. Cities offer many opportunities for innovative approaches to restructuring existing infrastructures. Therefore, creating sustainable cities for the future is essential in facilitating the necessary changes to combat the climate crisis.

All codes developed are sorted by chapter and can be accessed on Github:
https://github.com/LouisaHil/V2G-coupled-with-Urban-mobility-in-New-York

1.1 Background Information and Introduction

Fortunately, there have been significant advancements in recent years thanks to the quick adoption of innovative technologies. The 2022 IPCC Report [2] on climate change provides a thorough technical analysis of the progress made in net-zero energy systems and highlights the current challenges in implementing them, emphasizing the need for further innovation and systematic changes [2]. Our research will focus on integrating renewable energy sources into the existing infrastructure of New York City.

The IPCC report [2] noted significant cost reductions from 2015 to 2020 in important energy systems mitigation options such as photovoltaics (PV), wind power, and batteries. Specifically, PV, wind energy, and batteries experienced declines of 56%, 45%, and 64% [2], respectively, increasing both the capacity and generation of PV and wind energy. Solar PV grew by 170%, reaching 680 TWh, while wind energy expanded by 70%, totaling 1420 TWh [2]. These developments underscore the rapid adoption of renewable energy sources in recent years.

New York is ambitiously aiming for a clean energy transformation, targeting 70% of its electricity supply from renewable sources by 2030, doubling the current 31% [3]. By 2040, the goal is 100% [3]. To make this possible, New York is constructing an offshore wind farm with a capacity of 4,350 MW that will largely contribute to the 9,000 MW set goal by 2035 [4]. Currently, hydroelectric power provides about 23% of the state's electricity, whereas wind and solar will need to account for at least 45% to meet the 2030 goals [3]. If nuclear plants are not relicensed, wind and solar would have to supply about 67% of the electricity by 2040 [3]. Due to the unreliable

nature of wind and solar energy, the cost of this transformation includes the overbuilding of wind and solar energy facilities, substantial storage or backup energy production, and increased transmission costs [3]. Furthermore, the efficient use of renewable energy sources such as wind and solar power adds additional geographical and logistical challenges due to their remote locations, necessitating better transmission [5]. Additionally, with the growing electricity demand given by widespread electrification, these transmission lines must increase their capacity to distribute the necessary power. However, the power grid in the US has predominantly been based on a central station model, typically fueled by natural gas or hydropower, and has seen little significant change over the past century [5]. This requires grid reinforcement to facilitate the integration of distributed renewable energy sources. Achieving this while ensuring reliability is a complex task due to the grid's current centralized design and transmission constraints. The Clean Energy Leadership Act of New York State sets a goal for 3,000 MW of battery storage by 2030 to help maintain grid stability [6]. However, the economic feasibility of this approach is hindered by high costs and the growing demand for raw materials needed for batteries. Battery storage costs must be reduced by approximately 90% to make them financially competitive with fossil fuels [6].

To achieve this ambitious plan, New York must enhance its grid design, build more infrastructure, and introduce flexible resources to balance load with supply. This calls for carefully assessing energy storage solutions and robust transmission of high loads in the power grid.

Various energy storage systems exist, including flow batteries [7], compressed air energy storage [8], pumped hydroelectric storage [9], thermal energy storage [10], etc. that show potential solutions, but come with challenges. Each system has unique advantages and disadvantages in cost, scalability, lifespan, and efficiency. However, none of the latter energy storage systems are mobile and do not contribute significantly to the decentralization of the energy storage grid. Integrating mobility and decentralization in energy storage systems delivers key advantages, including enhanced grid resilience and improved efficiency by supporting dynamic demand response. Ultimately, shifting towards mobile, decentralized energy storage systems could lead to significant cost savings and a more robust, responsive, and equitable energy grid that facilitates the integration of renewable energy.

Vehicle to Grid (V2G) technology [11] is emerging and proposes a promising alternative to tackle these issues. V2G has existed for some time but has yet to be implemented on a large scale due to infrastructure constraints and a lack of economic incentives. V2G systems allow electric vehicles (EVs) to return energy to the grid during peak demand periods or when renewable energy production is low, providing a secondary utility beyond mobility. The potential of V2G technology has been the subject of much discussion, largely due to technical limitations, such as the possibility of reducing battery life. Although, some researchers suggest that under certain conditions, feeding energy back into the grid can potentially prolong the lifespan of batteries [12]. Additionally, implementing V2G technology would require a significant investment in bidirectional charging infrastructure. This includes high-speed DC charging stations with rates between 50-250 kW [13]. Currently, most charging stations operate at Level 1 and may not support bidirectional power flow [14]. Significant EV participation in New York would necessitate a corresponding increase in charging stations, which must align with the local grid's capacity. The grid's capacity determines the power that can be drawn for EV charging and the amount that can be returned to the grid via V2G. Excessive demand could lead to power outages or infrastructure damage. Furthermore, other studies have concentrated on the socio-economic difficulties of integrating V2G, with private vehicles assuming a role beyond mobility. Encouraging electric vehicle owners with economic incentives to charge and discharge their vehicles at specific times is a key factor in adopting vehicle-to-grid integration. Consequently, understanding and controlling vehicle mobility within the city's dynamic nature becomes even more critical.

In New York City, the city's energy and transportation systems are intertwined, and the increasing use of electric vehicles (EVs) highlights the need for a sophisticated approach to couple these. As EVs could account for 50-60% of transportation in the future [15], implementing vehicle-to-grid (V2G) technology can be transformative. Hence, our research centers around leveraging the mobility patterns of vehicles in New York City and coupling them with wind power generation to deduct the true technical potential of V2G.

The structure of this thesis begins with an outline of the methodology employed to simulate mobility patterns and the power generation of wind farms. Following this, a preliminary strategy is presented to interconnect mobility, the availability of wind power, and the charging and discharging activities of electric vehicles (EVs). These findings lay the groundwork for developing an innovative smart-charging algorithm, delving into the underlying dynamics that rule over EV fleets participating in V2G. In the process, we identify system limitations to ensure simulations are realistic and within computational constraints. In the concluding part of the research, a refined version of the smart-charging algorithm is presented, designed to calculate the actual number of EVs that participate in V2G when offshore wind power sources are integrated.

1.2 Literature Review on V2G and Renewable Energy Integration at the City Scale

Schläpfer et al. (2021) [16] presented a model that leverages anonymized mobile location data to deduce population-wide mobility and incorporates a vehicle charging/discharging scheme. This approach permits a refined V2G energy supply/demand assessment, enabling better photovoltaics deployment and infrastructure planning. The study emphasizes the importance of understanding human mobility patterns and urban transportation developments for effective V2G infrastructure planning.

Similarly, Xu et al. (2018) [17] utilized data from mobile phone activity, census records, and PEV charging behaviors to investigate the peak activity times and locations of PEV drivers. Their charging pattern analysis identified periods of high power demand that could potentially stress the power grid. To address this issue, they recommended a modified approach to PEV charging that accounts for the travel needs of individuals. The study's results indicated that implementing these changes could reduce peak power demand by up to 47%.

Ghofrani et al. (2014) [18] suggested using a stochastic framework to manage the unpredictable nature of wind power. The researchers proposed using vehicle-to-grid (V2G) capabilities (EVs) to improve wind power prediction and reduce unpredictability. They achieved this by using an Auto Regressive Moving Average (ARMA) wind speed model and grouping EVs based on their daily driving habits. To minimize costs related to wind power imbalances and V2G expenses, the researchers utilized a Generic Algorithm (GA) in an optimization scheme. Despite the main focus of the study being economic cost optimization, the unique aspect of this research was the pairing of wind power and mobility patterns.

To align EV charging and discharging with renewable wind power production, we have developed an innovative deterministic approach that requires the development of new methods. In the upcoming chapters, we will provide a detailed description of our new perspective on assessing the potential of V2G in New York. It is crucial to acknowledge that our analysis in this thesis is focused solely on the temporal aspect.

Chapter 2

Offshore Wind Power and Mobility Patterns in NYC

In this chapter, we will concentrate our analysis on harnessing wind energy from an Offshore Wind farm in New York and utilizing Electrical vehicles as a power balancing tool between electricity production and NYC electricity demand, enabled by V2G technologies. In the subsequent sections, we will outline our methodology for modeling wind power generation from the offshore wind farm. Additionally, a section of our study will explore urban mobility dynamics in New York City, explicitly emphasizing taxi and private vehicles. We aim to develop an initial model for V2G implementation in taxi mobility, which will provide valuable insights into the potential of using taxis as a part of V2G services.

2.1 Power Balance in NYC

2.1.1 Offshore Wind Power Generation

Given the significant technological advancements, offshore wind energy has become a widely used renewable. As of 2022, installed global offshore wind capacity's cumulative capacity has reached 63 gigawatts and indicates continual annual growth [19]. Integrating large-scale wind energy into the power system presents numerous challenges due to its intermittent and unpredictable nature. Accurate wind power forecasting ensures stable power system operations and optimizes wind power integration. Our study focuses on the challenges of maintaining a consistent power balance. Thus, as part of our project, we determined the most suitable method for modeling wind power generation to achieve our goal.

We looked into different methods for wind power forecasts that we could apply to our study.

Wind Power Modelling: Monte Carlo

To gain a general understanding of wind power output, we initially developed a basic Monte Carlo model based on a probability transition matrix from a wind farm in Spain derived in [20]. In this case, the states are classified between high, mid, and low wind outputs with the state matrix $W = [0.0112, 0.1092, 0.2385, 0.4026, 0.5409, 0.8152, 0.7624, 0.9199]$ with a maximum capacity factor of 1MW. We scaled this model to suit our requirements of 9000MW and conducted simulations over 24 hours to determine the power outcomes. However, this method introduced significant uncertainty and needed to be more suitable due to its reliance on the states and probability matrix derived from measurements of a wind farm in Spain, which may not apply to other locations. For accurate modeling specific to our desired location, New York City, we

recognized the need to derive transition probabilities and states, requiring additional measurements. In conclusion, we won't be going further into the specifics of this model as it is irrelevant to our thesis. Still, it was an important exploration step in wind power modeling. Nonetheless, the Monte Carlo simulation code, written in Matlab, is available on GitHub for reference. Given the initial Monte Carlo model's limitations, we explored alternative approaches for modeling wind power in our study.

Wind Power Forecast Modelling

Forecast methods are usually categorized based on time horizon [21]. Generally, a short-term forecast, which covers a period of hours to days, is used to ensure power supply quality. Mid-long-term forecasts over weeks, months, and years are used for wind farm designs and annual power generation prediction, while ultra-short-term forecasts are used mainly for wind turbine control [21]. Accurate prediction of wind farm power output leads over a wide range of conditions such as meteorological conditions, wind turbine arrangement in farms, and turbine wakes (velocity deficits after kinetic energy is taken out of the fluid and transferred into mechanical and kinetic energy) [22], control of wind farms [23], etc. Concluding, wind power forecast requires a deep understanding of dynamic interactions among turbines, wind farms, and the atmospheric boundary layer, whose flow produces the kinetic energy that drives power production [23]. We've established quickly that any attempt at accurately predicting wind power output would require extensive resources, which are out of scope for this master thesis. However, given the appropriate resources and knowledge, this research gives us an idea of the feasibility of predicting wind power output over time.

Continuing, we found significant data on the power output generation from offshore wind farms in New York City. The tool used is presented on the website https://renewables.ninja/. Renewables.ninja is a platform that simulates hourly power output from wind and solar power plants anywhere around the globe. The system uses weather data from global reanalysis models and satellite observations from [24], [25], and [26]. It uses the VWF model created by Iain Staffell [1], where wind speeds are converted into power output. This provides valuable insights for renewable energy analysis and planning. To extract the data, we chose the following parameters:

- Location: Long Island offshore
- Dataset: MERRA-2 (Global)
- Year: 2019
- Capacity: 9000MW
- Hub height (150m)- chosen as the predicted hub height of turbines in 2035 [27]
- Turbine model: Vestas V90 2000

The model generates hourly power output and wind speed data for the entire year of 2019. We need to process the data further to immediately have an overview of the capacity factor throughout the seasons shown in Fig.(2.1). We can observe seasonal discrepancies for very low power output in spring and summer (May through September) with a minimal capacity factor in August of 29% and high wind outputs in fall and winter (September through April) with a maximum capacity factor in January of 60.44%. We observe a total mean capacity factor of 48.3% over the entire year. The capacity factor gives a unit that presents the ratio between the actual energy generated and the theoretical maximum energy capacity [28]. In August, we would observe a minimum actual energy generation of 2610MW. The raw data download from [1] requires additional data processing to ensure that it fits our model. We set the local

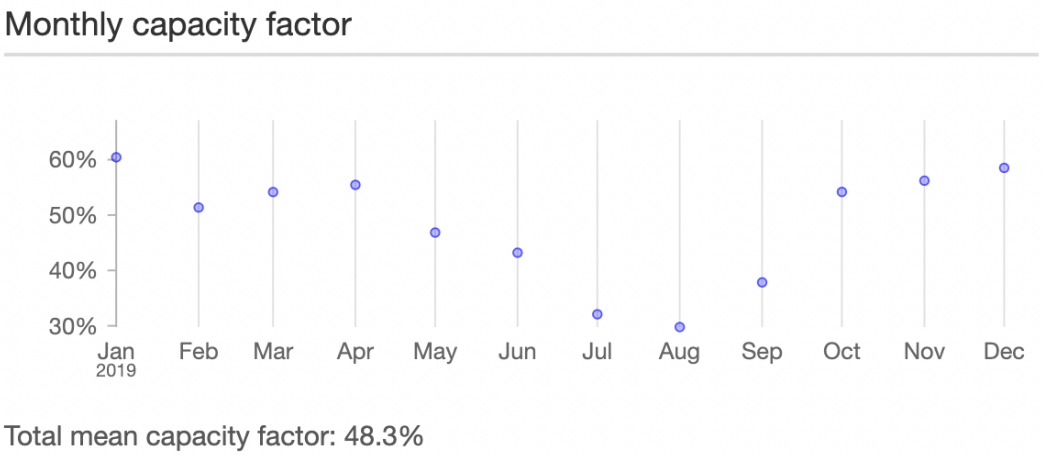


Figure 2.1: Capacity factor over entire year of 2019 [1]

time in New York City as our traceable time index and filtered for duplicates and missing values. We also ensure power output values in MW and interpolate the data to get a value every 15 minutes. The data processing code can be found on GitHub.

2.1.2 New York City Electricity Demand

Moving forward, we will now focus on creating a model for forecasting electricity demand in New York City. We acquired historical data on electricity demand for the NYC region from the New York Independent System Operator website NYISO [29]. The process of obtaining this data was easy and accessible. We obtained yearly load data for 2022 and processed it by removing duplicate entries, filling in missing values, and collecting corresponding load readings every 15 minutes. The codes utilized for data processing are available on GitHub for future reference.

After analyzing the data, we discovered that the minimum load recorded was 3853MW on May 2, 2022, at 05:00, while the maximum load was 9998MW at 12:30 on July 21, 2022. Generally, the distribution of electricity load exhibited a relatively uniform pattern, with peak demand during the day and lower demand at night.

2.1.3 Offshore Wind Power Generation vs. Electricity Demand

Our analysis compares wind power output with NYC's electricity demand, considering periods of excess and deficiency in power generation. This analysis determines the need for energy storage during surplus wind power and alternative sources of electricity production (or inflow power from storage) during insufficient wind power.

To highlight the significance of these considerations, we will first demonstrate that meeting NYC's demand solely with an offshore wind farm, without factoring in storage or alternative power generation, would require an unrealistically high capacity. The minimum wind power output would have to match the maximum NYC demand, which means an offshore wind farm capacity of about 140 000 MW which is an impractical scenario.

From Fig.(2.1), we can already say that we will not have coverage of power production over the entire month in June, July, and August for a capacity of 9000MW, and that we will require extensive discharging from V2G to meet NYC demand. For the other months, we might have

a power balance. In the following figures in Fig.(2.2), we have highlighted the surplus Power from wind output in green and the deficient Power in grey. Overall, we observe that for most months, we mostly have more deficient energy (Gwh) than excess energy, meaning even considering perfect storage solutions, additional power generation must most likely be provided. We have accumulated enough power production from wind offshore farms over the entire month for April, October, and November. With appropriate storage solutions, a wind offshore farm with a capacity of 9000MW could sustain NYC electricity demand. However, over the periods of the other months, we do not provide sufficient energy over the months to cover all NYC demand. Additionally, we ran our analysis with an offshore wind farm of 13000MW for comparison. In Fig.(2.3), we have depicted the cumulative excess and deficient energy for all months for a capacity of 9000MW and 13000MW. Even with a capacity of 13000MW, we would not have sufficient accumulated energy production for July, August, and September. We expect the easiest integration of energy storage to balance out the power grid in months with a small positive offset between excess and deficient energy.

The optimal scenarios for energy storage integration would therefore be for the following:

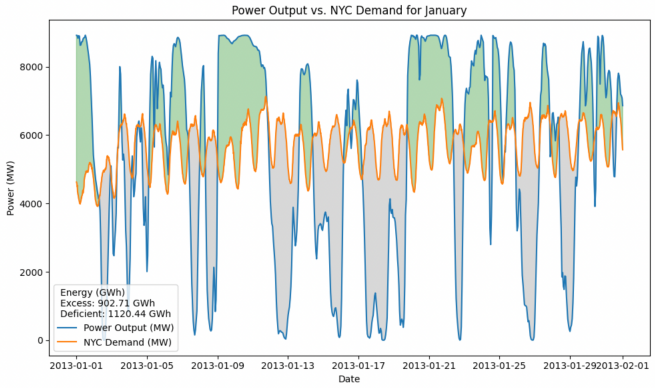
- 9000MW in April(+227.27 Gwh), October(+149.93 Gwh) and November(+39.13 Gwh)
- 13000MW in June (offset: +58.9 Gwh)

We will also consider months with a small negative offset to asses the feasibility :

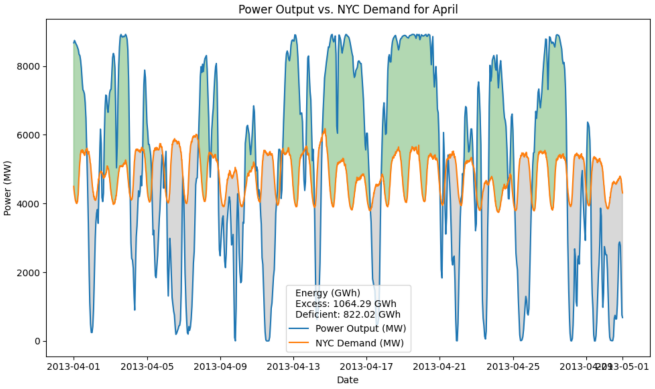
- 9000MW in March(-22.68 Gwh), December (-100.06 Gwh)

For critical cases, we would like to consider the following cases.

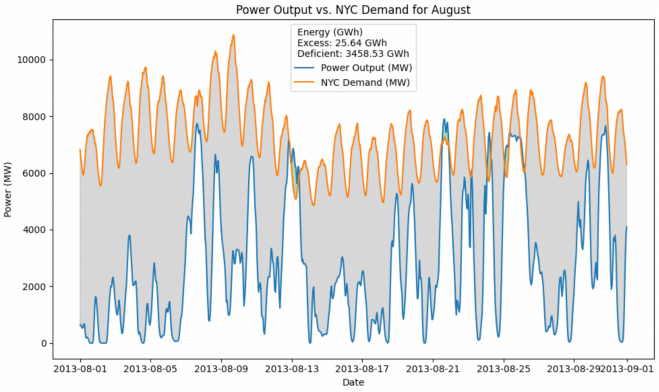
- Very low offset: 9000MW in August (-3432.89 GWh)
- Very high Offset: 13000MW in January(+1850.55 Gwh)



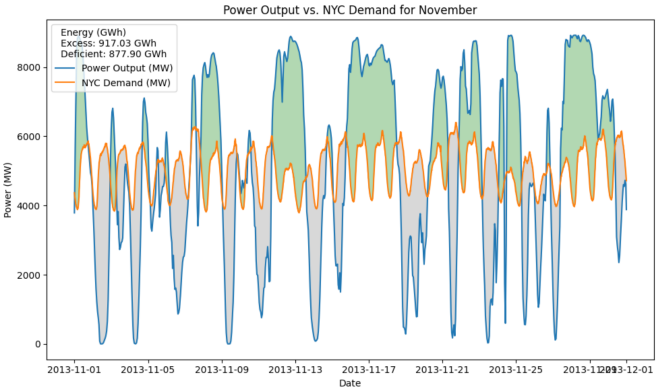
(a) Wind power generation vs NYC electricity demand in January



(b) Wind power generation vs NYC electricity demand in April



(c) Wind power generation vs electricity demand in August



(d) Wind power generation vs NYC electricity demand in November

Figure 2.2: Wind power generation vs. NYC electricity demand

9000 MW	1	2	3	4	5	6	7	8	9	10	11	12
+ GWh	902.71	677.79	1006.61	1064.29	782.45	346.94	4.88	25.64	176.48	1087.27	917.03	899.40
- GWh	1120.44	1224.70	1029.29	822.02	1296.45	1678.67	3221.31	3458.53	1848.59	937.27	877.90	1000.06

13000 MW	1	2	3	4	5	6	7	8	9	10	11	12
+ GWh	2528.95	1820.8	2368.2	2501.2	1910.1	1335.6	288.2	316.4	926.8	2492.8	2425.75	2451.12
- GWh	678.40	826.5	736.3	694.8	921.8	1296.07	2342.8	2457.5	1270.4	701.98	560.8	695.80

Figure 2.3: Energy deficiency and excess for wind power capacity of 13000MW and 9000MW against NYC demand over all months (1-12)

2.2 Mobility Patterns in New York City

Next, we will focus on deriving New York City's vehicle mobility patterns to introduce the integration of Vehicle-to-Grid technology (V2G) driven by electric vehicles (EVs). We aim to understand when energy storage from EVs would be available and if the charging and discharging processes can be aligned with wind dynamics to maintain the power balance. This section focuses on two key vehicle types: taxis and private vehicles. Studying taxi trajectories offers valuable insights into general mobility behavior in New York City. It serves as the basis for realistic assumptions in deriving a mobility model for private vehicles. Furthermore, although buses and carsharing vehicles equally have a big potential for V2G, their limited data availability led us to concentrate our analysis on taxis and private vehicles. The following mobility analysis assumes all car trajectories are done with electric vehicles.

By carefully assessing mobility patterns, we aim to facilitate the implementation of efficient Vehicle-to-Grid (V2G) solutions and sustainable power grid balancing, taking into account the unique characteristics of taxi and private vehicle usage in New York City.

2.2.1 Taxi Fleets

To evaluate the driving and parking habits of taxi vehicles, we refer to the Taxi Limousine Commission (TLC) dataset from 2013 [30]. According to the law, 13587 taxis are operating in New York City. The dataset from 2013 contains information on 13,587 yellow and green medallion cabs, including trip records such as pick-up and drop-off dates and times, locations, distance traveled, fare details, rate types, payment methods, and number of passengers. Typically, trips within Manhattan are short (around 1.9 miles) and quick (averaging 10 minutes). While the TLC does not regulate specific shifts, the morning shift for a medallion taxi typically begins at 5:30 a.m., and the evening shift often starts at 5:15 p.m. Trips peak Friday evenings for medallion cabs and Saturday night for green taxis. From our dataset of 2013 from the TLC, we have information on the specific trajectory of a taxi with a corresponding medaillon number over the entire year. The datasets are split by month, so the evaluation of trajectories will be separate for every month.

Data Processing

From our data, we extract the relevant columns : Medaillon Nb, Pickup datetime, dropoff datetime, trip distances, pickup, and dropoff location coordinates. We will base our analysis on the temporal aspect of taxi mobilities. We can group the taxi trajectories by medallion number

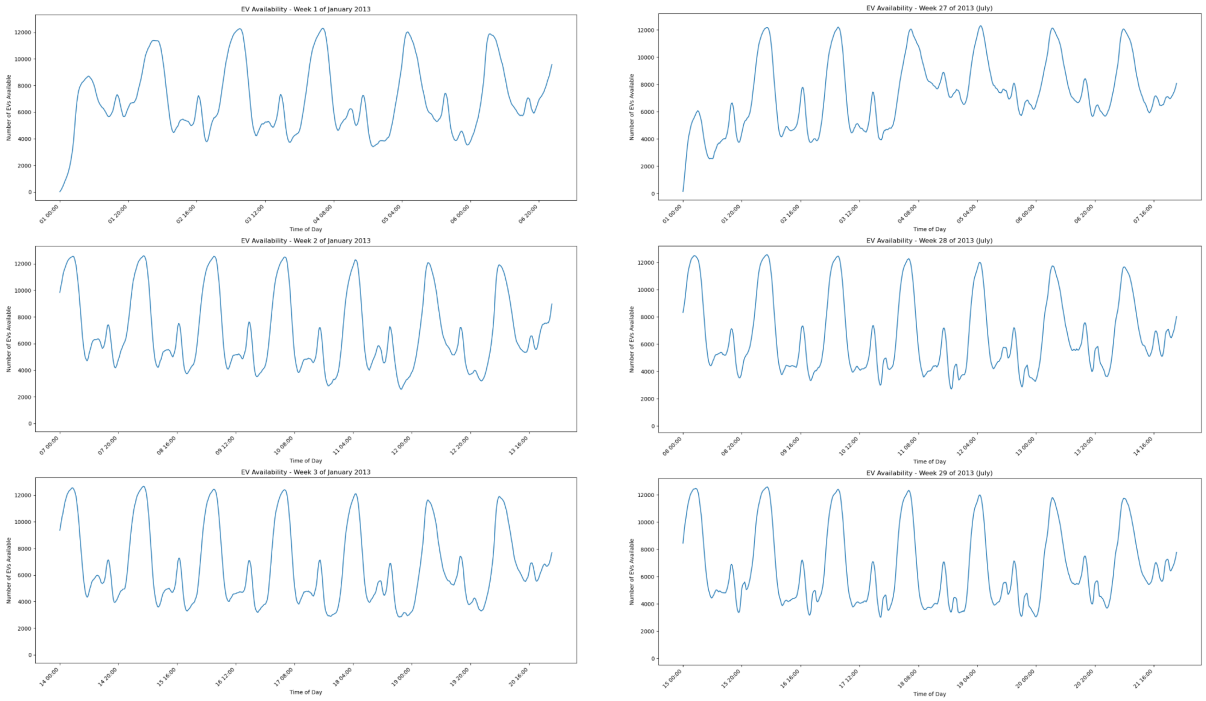
and then track the off (not on a trip) and on times (on a trip). If the off time is longer than 1 hour, we assume the car is available for charging and discharging. We will present a small pseudo-code explaining our methodology for processing the data from TLC.

Pseudo Code

We create an availability matrix with 0 as driving and one as parked with a datetime index starting at 2013-month-01 00:00 and ending at 2013-month-31 23:45 with values for every 15 minutes. Next, we loop over every medaillon and extract the off-times (when a taxi is not performing a trip); we check if this off-time is longer than 1 hour and then account for parking in this off-period. Next, we overlap the off-periods of each medallion with the indexes created in our availability matrix and fill the availability matrix if, within this 15-minute interval, the taxi is parked (1) or driving (0). This analysis allows us to identify parked and driving times for every medaillon over an entire month. Additionally, by taking the sum at each time index of the availability matrix over all taxi medallions, we can identify the total number of taxis parked in NYC at each time.

Methods and Results

Mobility patterns in New York City seem periodic throughout the year, with no real seasonal discrepancies. Taxis are generally mostly parked between 00:00-07:00 AM and operating between 9:00 AM-1:00 PM and 7:00 PM and 10:00 PM. To demonstrate our analysis, We plotted the mobility patterns in Fig.(2.4) over randomly chosen weeks in January and July. We reach a maximum number of available (parked) cars of 12000 and a minimal number of about 4000 taxis.



(a) Weekly Taxi availability in January

(b) Weekly Taxi availability in July

Figure 2.4: Weekly Taxi availability patterns

2.2.2 Private Vehicles

Additionally, we would like to extract the trajectories of private vehicles over a year. We considered using simulation tools such as MATSIM (NYC) [31] to determine the mobility patterns in New York City. MATSIM is an open-source transportation model that covers the entire New York City area. It incorporates the latest technologies, such as connected and autonomous vehicles, electric vehicles, dockless bikes, and ridesharing systems, into its simulation. By combining data and modeling capabilities, it offers a comprehensive overview of transportation in the city [31]. As MATSIM (NYC) generates simulated data and our approach is deterministic, it will not be practical for our use case. We have used the NHTS (National Household Travel Survey) dataset instead [32]. This dataset contains information on vehicle trips per household on a specific day, which is the basis for the MATSIM (NYC) transport simulation. The 2010/2011 National Household Travel Survey (NHTS) is a valuable resource for travel information in any region. It covers 143,925 trips from 18,965 households and 43,558 participants. Moreover, 1,930 households provided travel data using wearable Global Positioning System (GPS) devices. The survey provides data for all types of transport: car, rideshare, bike, bus, taxi, etc. From the dataset: unlinked trip data in [32], we filtered by transportation mode, and we were able to extract:

- Car ID (household number + vehicle number)
- Arrival and departure time
- Weekday
- Trip duration and distance
- Motive of trip

Noticeably, we need to get information on the exact date and time of the month from this dataset. We only have data on arrival and departure times and the weekday (Monday through Friday). Keeping the goal in mind that we would like to couple mobility patterns to the wind dynamics from a temporal perspective, we need to derive a time series spanning over the integrality of the month with a frequency of 15 minutes. Hence, we will compute the distribution of arrival and departure times over a week from the NHS dataset.

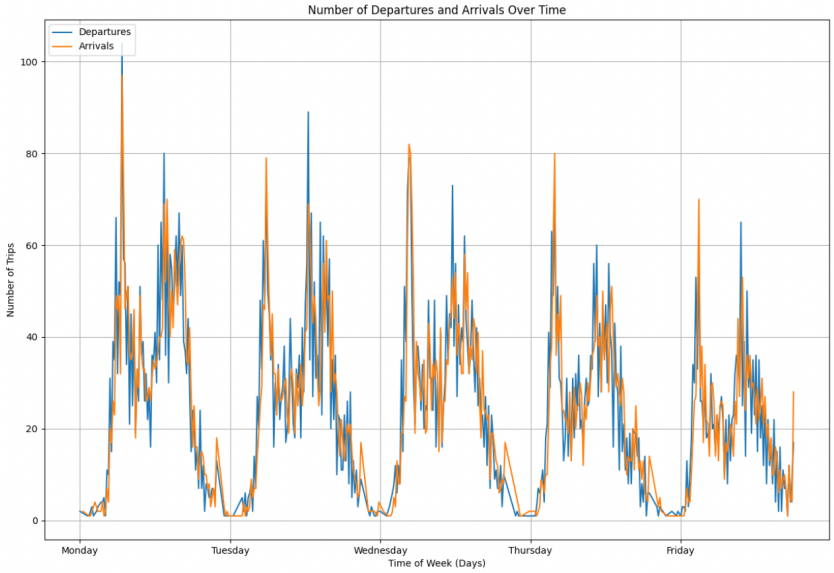


Figure 2.5: Departure and arrival times of individual trips over a week

We can observe that the patterns over the week are uniform, and we observe two daily peaks: one in the morning between 7:00 AM and 9:00 AM and one in the afternoon between 5:00 PM and 8:00 PM. As we have seen, taxi mobility patterns do not vary significantly over the entire year and are not coupled to seasons. We will therefore assume that we can replicate the weekly mobility patterns over the entire year.

Data Processing

The code for the preprocessing of the dataset can be found under travelsurvey.py on GitHub. The code processes travel data, filters relevant information, performs data transformations and calculations, and produces a modified dataset for further analysis.
Pseudo code:

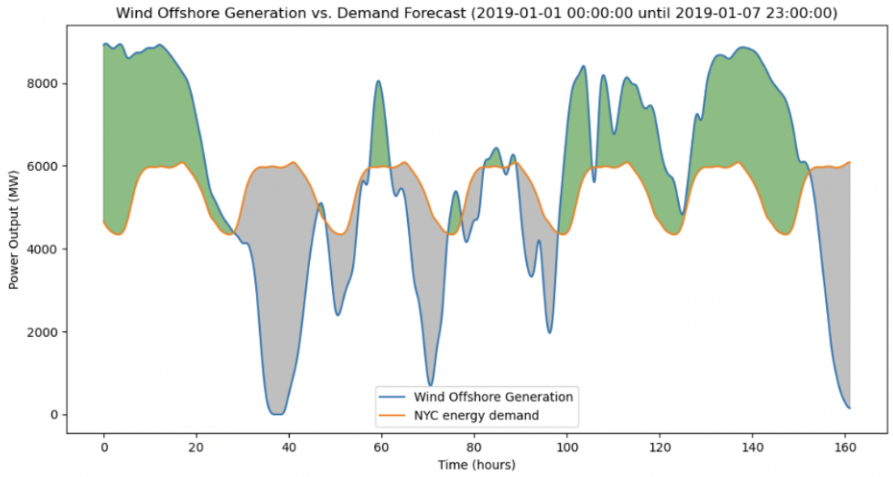
- Data is filtered by city (New York City) and mode of transportation (vehicle)
- Data manipulation tasks: adding a date to time columns, duplicating entries for each day, grouping data by vehicle ID
- Output: Trip data (monthly)
- Calculating the number of parked cars during specific time intervals
- Calculating availability by generating time intervals, initializing an availability matrix, and iterating over unique car Ids. It considers off-periods (parking), identifies overlapping time intervals and updates the availability matrix accordingly.
- Output: expanded availability matrix (yearly and monthly)

From the NHTS, we only obtain 2964 different car trajectories (availability matrix size) over an entire year. However, as previously shown, the mobility patterns are uniformly distributed over the day. Therefore we can generate further car trajectories by computing the driving and parking probabilities at each time instance and then using a binomial distribution to generate more data. The codes for scaling the availability matrix to a higher count of cars can be found on Github under Scalingmobility.py.

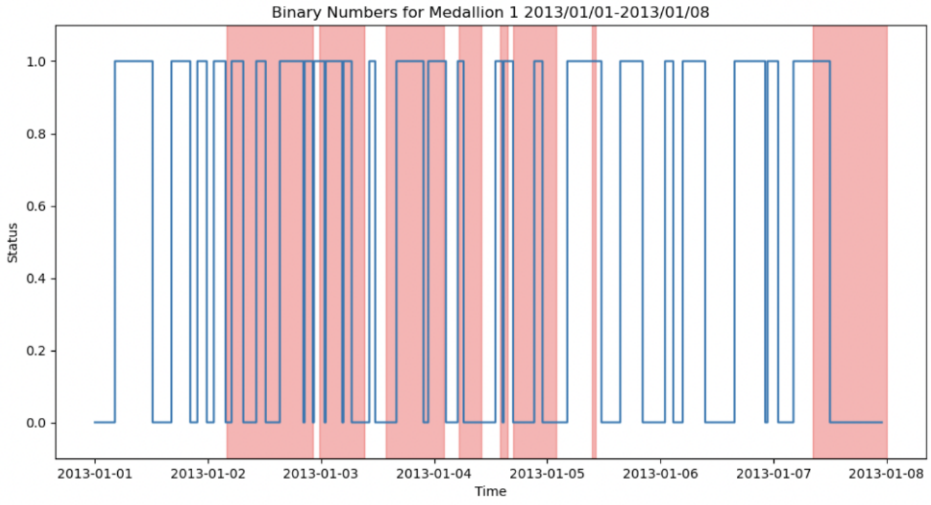
2.3 V2G Technology: Assessment of State of Charge (SOC) of EV

2.3.1 Coupling of Taxi Mobility and Wind Power Availability

In the first instance, we are mainly interested in comparing the temporal mobility patterns to the temporal dynamics of wind power. We will overlap the periods of wind insufficiency with the parking availabilities of taxis. Ideally, the taxi can discharge its battery in these periods to feed power back to the grid. As previously determined, the patterns of wind energy could be more coherent, meaning a simple rule to couple them with mobility patterns will not be successful. For this reason, we determine a deterministic model where we evaluate at each time step wind availability/deficiency and asses the total amount of power given back to the grid by the available taxis. In the following plots, we have visualized the mobility patterns of taxis against low wind periods for a randomly chosen week in January.



(a)



(b)

Figure 2.6: Wind Offshore generation for the 1st week of January and Taxi mobility with low-wind periods marked in red for the first week of January. Status: 1- parked, 0-driving

To evaluate how much power can be discharged, we need to have the following information available:

- Wind power availability
- Mobility availability
- Variable parameters: SOC_{min}, SOC_{mid}, charging rate λ_c and discharging rate λ_d

And we are interested in computing :

- State of charge (SOC) levels
- Time available for discharging
- Total Power discharged from EVs

The variables SOC_{min} and SOC_{mid} are used as constraints for discharging and charging. To perform this analysis, we must introduce a crucial component for determining the potential: the State of Charge of the Battery (SOC).

State of Charge Assessment (SOC)

"The state of charge of an electric vehicle is defined as the ratio of the available capacity and the maximum possible charge that can be stored in a battery" [33]. Therefore, we need to ensure that at every timestep, a car has a high enough state of charge to complete its future trip.

System Definition:
We only discharge a car when the wind is low, and the SOC is not below SOC_{min}. At every other time instance, we charge a car. The choice of the different variables is crucial and depends on many aspects. The SOC_{min} will give a constraint on when we decide not to charge anymore. Choosing the optimal parameters will depend on various factors, such as:

- The driving patterns and habits of the EV owner: If the owner has a predictable daily commute, a lower minimum SOC might be sufficient. On the other hand, if the owner has irregular or longer trips, a higher minimum SOC might be necessary.
- The capacity and efficiency of the EV battery: Larger battery capacities or higher efficiencies may allow for a lower minimum SOC, as they can store more energy and cover longer distances with a smaller percentage of charge.
- Charging and discharging rates- have a significant impact as they determine the speed of charging and discharging

The optimal parameters are different for every car and cannot be chosen constant for all cars. We will therefore implement an algorithm computing the SOC at each time t and simultaneously find the optimal combination of parameters to optimize charging/discharging EVs based on wind availability for every taxi medallion.

Scenario Description

Firstly, we need to define fixed parameters:

- Maximum Battery capacity: C_b =70 kWh
- Energy consumption: E_c= 0.34 kWh/miles
- Average speed: v= 8.8 miles/hour
- Parking or driving time (h): Δt=0.25
- Total amount of energy used during trip: ΔE_u=E_c*v*Δt=0.75.

Secondly, we need to define all variable parameters. They differ for each EV but are constant over time. In our algorithm, the selection of these variables will vary as each parameter is given a realistic range. We introduced the ranges as follows:

- Discharging rate: λ_d =70-20 kW with an increment of -10
- SOC_{min}: 0.3-1 with an increment of 0.05
- SOC_{mid}: 0.65-1 with an increment of 0.05

Ultimately, we will assess the different constraints for each taxi and then decide on what action to take. The decision scheme is represented in Fig.(2.7).

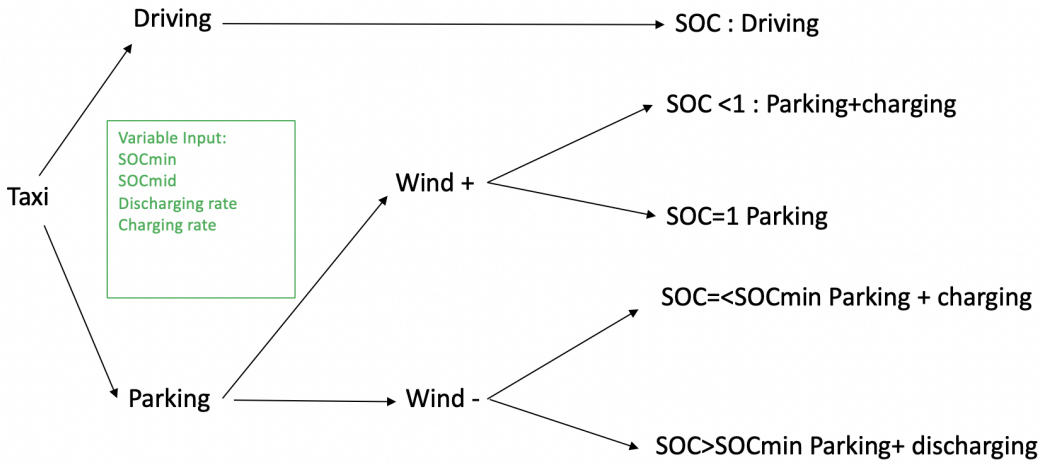


Figure 2.7: Charging-discharging procedure

We assess for every t in $[t_k, t_{k+n}]$ where n is the number of changes in driving behavior (driving-parking). Looking at our plot in Fig.(2.6), it is each time we have a jump in our step function. k is a number from 0 to n. We need to differentiate between three different periods:

- Driving
- Periods of parking and excess wind power generation
- Periods of parking and insufficient wind power generation

The SOC of each taxi is updated according to the following scheme:
Driving

$$SOC(t_k) = SOC(t_{k-1}) - \frac{\Delta E_u}{C_b} \tag{2.1}$$

Periods of parking and excess wind power generation
-Parking and charging with Δt: park duration

$$SOC(t_k) = SOC(t_{k-1}) + \min\left\{\frac{\lambda_c \cdot \Delta t}{C_b}, 1 - SOC(t_{k-1})\right\} \tag{2.2}$$

Periods of parking and insufficient wind power generation
- if $SOC(t_{k-1}) \geq SOC_{min}$, then parking and discharging with Δt : Park duration.

$$SOC(t_k) = SOC(t_{k-1}) - \min\left\{\frac{\lambda_d \cdot \Delta t}{C_b}, SOC(t_{k-1}) - SOC_{min}\right\} \tag{2.3}$$

if $\frac{\lambda_d \cdot \Delta t}{C_b} > SOC(t_{k-1}) - SOC_{min}$:
compute the new duration of discharge :

$$T_d = C_b * \frac{(SOC(t_{k-1}) - SOC_{min})}{\lambda_d} \tag{2.4}$$

- if $SOC(t_{k-1}) < SOC_{min}$ then parking and charging until SOC_{mid} is reached. SOC_{mid} is introduced as we would like to limit the amount of charging during these periods as no wind power is available.

$$SOC(t_k) = SOC(t_{k-1}) + \min\left\{\frac{\lambda_c \cdot \Delta t}{C_b}, SOC_{mid} - SOC(t_{k-1})\right\} \tag{2.5}$$

if $\frac{\lambda_c \cdot \Delta t}{C_b} > SOC_{mid} - SOC(t_{k-1})$ then compute T_c to reach SOC_{mid}.

$$T_c = C_b * \frac{(SOC(t_{k-1}) - SOC_{mid})}{\lambda_c} \tag{2.6}$$

We run our algorithm over a month and assess the correctness of our scheduling algorithm: SOC can never fall below zero, and the energy level in the battery is never above C_b. We evaluate the new combination of parameters for each failing scheduling algorithm until the system is stabilized. The order of new variable assessment is chosen given that the goal is to have the maximum discharging time and high discharging rate to increase the energy available that can be given back to the grid: So we choose our constraints and prioritize them in order with:

- SOC_{min} should ideally be low (0.3-0.8)
- SOC_{mid} should be low (0.5-1)
- r_d high
- r_c low or high (technical constraint)

We compute a car's charge time and discharge at every instance t_k in low wind periods. From this, we can compute the energy in every time interval taken (charging) or given to the grid (discharging) by multiplying the time of charge/discharge with the respective charging(-) and discharging rate(+) of the car id. We take the sum over all car IDs at each time index and divide it by Δt to obtain the actual power from discharging/charging cars in periods of low wind.

2.3.2 Results and Discussion

We ran the algorithm for taxis and private vehicles and computed the resulting power from discharging cars in low wind periods.

Taxi Fleet

Based on our observations, taxis have a low impact on power generation as they are primarily in operation with very few non-driving periods. However, we are limited by the number of taxis operating in New York, around 13000 cars. For this number of taxis, we have observed a maximum output of 495.49 MW during low wind periods (February) and an average output of 71.97 MW. Unfortunately, this amount of power generated from taxis is insufficient to cover the high demand for power in NYC. As shown in Fig.(2.8), it is clear that the power balance requirement (orange curve) cannot be achieved with V2G procedures (blue curve). We have only assessed the power balance during low wind periods to understand the potential for discharging

from V2G. It is critical to consider periods of low wind as they impact the mobility of taxis, which is their most important utility.

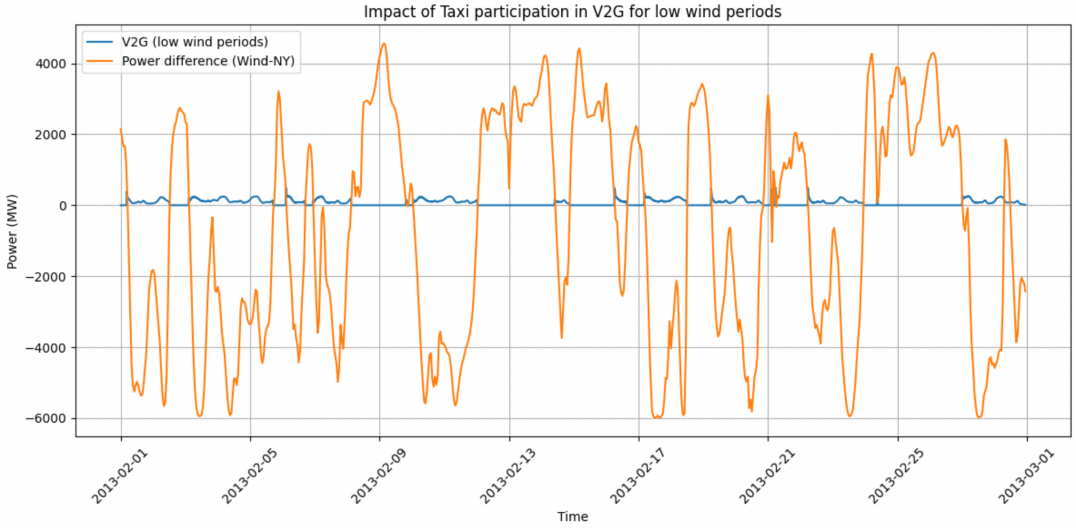


Figure 2.8: Impact of Taxi participation in V2G for low wind periods

Private Vehicles

We are running calculations for private vehicle trajectories, with the expectation that parking will be a more likely occurrence than driving. Due to a limited dataset of only 2694 cars, we need to scale up the power calculated. To achieve this, we will use a simple model to compute the available power from 2694 and then scale it up to our desired EV fleet of 1,000,000 cars. As seen in Fig.(2.9), we can cover NYC demand during deficient wind power generation periods in January. As we previously discovered, we will need a much larger fleet size during months with significantly lower wind power generation, such as July, August, and September. Based on our analysis from January, we expect a fleet size of 1.0 to 2.0 million cars to sustain demand throughout the year. However, we also observe high overshoots of Power given back to the grid with high EV fleet participation.

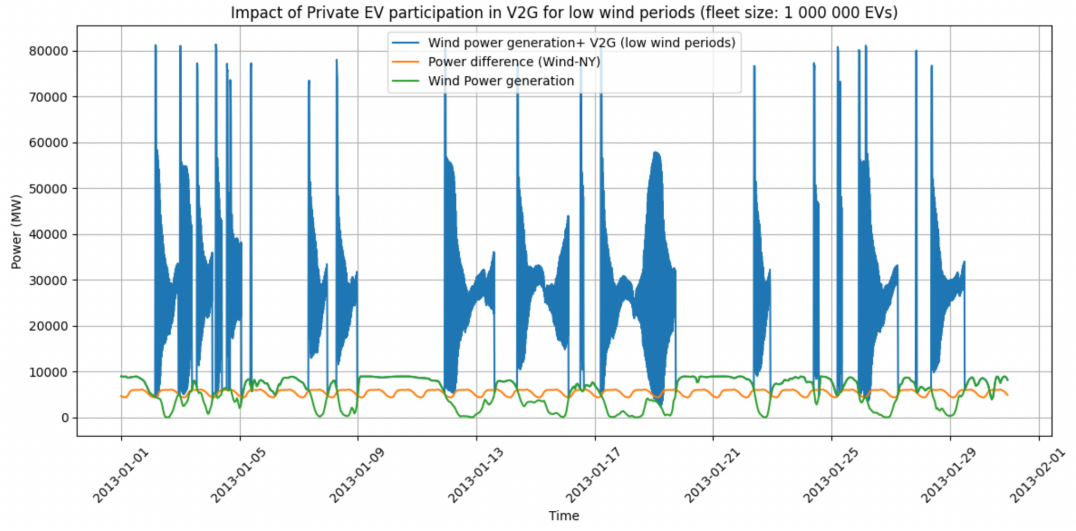


Figure 2.9: Impact of private electrical vehicle participation in V2G for low wind periods (fleet size: 1 000 000 EVs)

We must reassess our algorithm to reduce the overshoots while covering NYC electricity demand. We should reduce the fleet size and schedule discharging/ charging of each EV, so we do not overshoot. For example, after high wind and high car availability, switching into a period of low wind, almost all cars will start to discharge simultaneously. Similarly, after a long period of low wind and during peak hours, we will require a much bigger fleet to sustain demand. The formulation of this problem adds other factors to our problem: At every instance t, we need to take into account the power difference ($\mathrm{P}_{\mathrm{diff}}(t)$) and charge/discharge just as much that the power balance is always satisfied. This condition adds the following constraint to our system:

$$\mathbf{P}(t) = P_{\mathrm{V2G}}(t) + \mathrm{P}_{\mathrm{diff}}(t) \approx 0 \tag{2.7}$$

where :

$$\mathrm{P}_{\mathrm{diff}}(t) = P_{\mathrm{wind}}(t) - P_{\mathrm{NYC}}(t) \tag{2.8}$$

Naturally, it is essential to consider the same logic for periods of high wind when charging electric cars. Only as much power as excess energy can be used to charge cars. This will significantly affect the output, especially when determining the fleet size needed to meet the electricity demand in NYC at all times. Since there are more low wind periods than high wind periods, and we have limited charging available for the entire EV fleet, the fleet's overall state of charge (SOC) dynamics will be significantly impacted. We must accommodate larger fleets to participate in the V2G scheme. To address these limitations, we need to revise the algorithm and consider the individual dynamics of the SOC of each EV over time, as well as the general fleet dynamics of the SOC. Specifically, we need to account for the interdependencies of each EV in the fleet.

Chapter 3

Smart-Charging Algorithm

This chapter will give a thorough description of our novel Smart-scheduling Algorithm. Firstly, we will define the scenario at hand. Secondly, we will introduce all crucial variables and constraints of our methodology. This will lead to a detailed system analysis that will serve as a base for any alterations and control attempts to our algorithm. It will be a crucial part of our study to acquire the proper intuition into the underlying dynamics and bottlenecks this problem encounters. Lastly, we will discuss our results and present potential improvements to our method. For further assessment, all codes and data can be found on GitHub.

3.1 Methodology

3.1.1 System Definition and Constraints

In this section, we will outline the key characteristics and constraints of our SCA algorithm and identify the critical variables that determine the results of our model. This description is designed to help define the scope of our model and lay out the operating rules within this context.

The following variables define the system:

- Time interval (Hours):Δt
- Battery capacity (kWh): C_b
- Energy loss from driving (kWh): ΔD
- Charging rate (kW): λ_c
- Discharging rate (kW): λ_d
- Offshore wind power output capacity(MW): C_W
- Scaling factor: α
- Size of EV fleet from dataset: N

We have the following data available:

- Parking/Driving availability
- wind power output vs NYC electricity demand

These variables serve as the backbone of our model, playing a crucial role in defining the operational dynamics and constraints for the electric vehicles (EVs) within our system.

Scenario Description

The scenario under consideration involves a fleet of N EVs in New York City. Information about the parking and driving schedules of each EV is available at intervals of 15 minutes. Furthermore, the dataset contains wind power generation records from New York City and NYC power demand, recorded at 15-minute intervals throughout the year.

Let $\mathrm{P}_{\mathrm{diff}}(t)$ denote the power difference between wind power generation and NYC power demand at a given time t, given by

$$\mathrm{P}_{\mathrm{diff}}(t) = P_{\mathrm{wind}}(t) - P_{\mathrm{NY}}(t) \tag{3.1}$$

The objective is to optimize the quantity of active EVs engaged in charging or discharging at a given time, to minimize the power mismatch defined as $\mathbf{P}(t)$:

$$\mathbf{P}(t) = P_{\mathrm{V2G}}(t) + \mathrm{P}_{\mathrm{diff}}(t) \tag{3.2}$$

where $P_{\mathrm{V2G}}(t)$ refers to the power contributed by Vehicle-to-Grid (V2G) services. This quantity is positive when the power is supplied to the grid and negative when power is drawn from it.

Each EV from the fleet can perform one of the following actions at a given time:

- Neutral
- Charging
- Discharging

Additionally, each EV can exist in one of four states:

- Driving
- Parked
- Critical
- Blocked

The action an EV undertakes is dependent on its current state, the actions and states of the other EVs in the fleet, and the value of $\mathrm{P}_{\mathrm{diff}}(t)$. The actions "charging" and "discharging" represent the EV's participation in the V2G program, during which it either supplies or receives power from the grid.
The objective of the study is to determine the distinct states and actions of each electric vehicle within 15-minute intervals, thereby enabling us to compute the power either extracted from or contributed back to the grid by these EVs at every time instance. The goal is to obtain the number of EVs needed to participate in the V2G scheme in New York to enable the integration of offshore wind power coupled with energy storage by Electrical vehicles.

The following section outlines the complexities of the Smart Charging Algorithm (SCA). We start by implementing the constraints that each vehicle is subjected to when selecting an action. Subsequently, we outline the calculations involved in determining $P_{\mathrm{V2G}}(t)$, which is contingent on the number of EVs participating in the V2G program at a given time. Lastly, we comprehensively assess the SCA's system dynamics by examining its interdependencies and reevaluating our initial assumptions, specifically about the choice of scaling factor.

3.2 Computations

3.2.1 Algorithm Definitions and Variables

In this segment, we will provide a detailed exposition of the computations performed during this study. For a more hands-on exploration of the calculations and the ensuing results, the reader is directed to the GitHub repository, where the associated codebase can be accessed and reproduced.

Nomenclature

The following set of equations establishes the nomenclature and variables used in the algorithm:

$$\begin{aligned} \text{W+, when: } & \mathrm{P}_{\text{diff}}(t) \geq 0 \\ \text{W-, when: } & \mathrm{P}_{\text{diff}}(t) \leq 0 \\ & D_{+}\text{, when charging} \\ & D_{-}\text{, when discharging} \end{aligned} \tag{3.3}$$

$$Set_cars(t) = \{i \in \{0, Ncars\}, \text{car}_i \text{ is a car in a set of N cars at time} t\} \tag{3.4}$$

$\forall i \in \text{Set_cars}(t), \text{Car}_i$ has the following attributes:
$\text{SOCmin}(t+1)_i$: the minimum SOC of car i,
$\text{SOC}(t+1)_i$: the SOC at time $t+1$ of car which is being calculated at every timestep t,
$\text{SOC}(t)_i$: the SOC at time t of car i,
i : the index of the car,
$\text{Power}(t)_i$: the power from discharge/charge of car i.

$$Set_cars_parked(t) = \{i \in Set_cars(t) : \text{Car}_i \text{ is parked}\} \tag{3.5}$$

$$Set_cars_driving(t) = \{i \in Set_cars(t) : \text{Car}_i \text{ is driving}\} \tag{3.6}$$

$$Set_cars_blocked(t = \begin{cases} i \in Set_cars_parked(t) : \text{SOC}(t)_i < \frac{\lambda_d * \Delta t}{C_b} & \text{if } W_{-} \\ i \in Set_cars_parked(t) : \text{SOC}(t)_i > 1 - \frac{\lambda_c * \Delta t}{C_b} & \text{if } W_{+} \end{cases} \tag{3.7}$$

$$\begin{aligned} Set_cars_critical = \{i \mid i \in Set_cars(t) \\ \cap Set_cars_driving(t+1) : \\ \text{SOC}(t)_i - \frac{\lambda_d * \Delta t}{C_b} \leq \text{SOCmin}(t+1)_i\} \end{aligned} \tag{3.8}$$

$$Set_cars_available(t) = \{i \in Set_cars_parked(t) : \text{Car}_i \text{ is parked,not critical and not blocked}\} \tag{3.9}$$

$$Set_cars_critical_available = \{i \in Set_cars_parked(t) : \text{Car}_i \text{ is parked, critical and not blocked}\} \tag{3.10}$$

Relationship between the sets (from set theory). This is useful to have in mind when implementing the code.

$$Set_cars_driving = Set_cars \setminus Set_cars_parked \tag{3.11}$$

$$Set_cars_available = Set_cars_parked \setminus (Set_cars_critical \cup Set_cars_blocked) \tag{3.12}$$

$$Set_cars_critical_available = Set_cars_critical \setminus Set_cars_blocked \tag{3.13}$$

3.2.2 SOC_{min} (Minimal State of Charge): Pre-Computation

The term SOC_{min} is crucial in the functioning of electric vehicles (EVs) as it represents the minimum state of charge. SOC_{min} is the lower limit of SOC that can be reached without causing any harm to the battery or reducing its lifespan [33].
For our purposes, we define the SOC_{min} as the lowest threshold at time t for which the car is still able to accomplish its upcoming trips. This will involve a backward computation of our mobility dataset. The algorithm starts from the end of each car's driving record and counts the number of consecutive drives. For each consecutive driving period, it reduces the SOC by the fraction of the energy consumed in that drive relative to the total battery capacity. We create a function to determine SOC_{min} by processing the driving history of each car in *Set_cars* in reverse order.
The algorithm is implemented as follows:
Let $d_i(t)$ represent whether the car i is driving (0) or parked (1) at time t:

$$d_i(t) = \begin{cases} 0, & \text{if car } i \text{ is driving at time } t \\ 1, & \text{if car } i \text{ is parked at time } t \end{cases} \tag{3.14}$$

To identify individual trips, we can compute the groups $G_i(t)$ with the formula:

$$G_i(t) = \sum_{s=t}^{T} [d_i(s) \neq d_i(s+1)] \tag{3.15}$$

In this equation, T represents the total number of time steps. The $[\cdot]$ notation is an indicator function that checks whether the condition inside it holds true. If $d_i(s)$ isn't equal to $d_i(s+1)$, it gives a 1 (a 'yes'), and if they're equal, it gives a 0 (a 'no'). Next, the cumulative sum of drives $D_i(t)$ is computed for each car i and group $G_i(t)$:

$$D_i(t) = \sum_{s=t}^{T} [d_i(s) == 1] \cdot G_i(s) \tag{3.16}$$

Finally, the minimum SOC is computed: $SOC_{\min,i}(t)$ for each car i at time t:

$$SOC_{\min,i}(t) = \max\left(0, \frac{D_i(t) \cdot \Delta D \cdot \delta t}{C_b}\right) \tag{3.17}$$

Lastly, the times are reversed in the correct order.
By pre−determining the SOC_{min} for each vehicle at each time step, the algorithm ensures that the selected actions (i.e., charging, discharging, or remaining neutral) for the vehicles will not let their batteries fall below the safe minimum level of charge. This computation is independent of the SCA and is done beforehand for all cars in *Set_cars* and is an attribute of the set.

3.2.3 SCA (Smart-Charging Algorithm): SOC and Power computations

After calculating the $\mathrm{SOC}_{\min}$, we will determine the SOC and power exchanged through V2G for each car individually. First, the power taken or given to the Power gird and the SOC of all cars that are driving are computed with the following equations:

$$\begin{cases} \mathrm{SOC}_i(t) = \mathrm{SOC}_i(t-1) - \dfrac{\Delta D * \Delta t}{C_b}, \\ \mathrm{Power}(t)_{i,\mathrm{driving}} = 0 \end{cases} \quad \forall i \in \text{Set_cars_driving} \tag{3.18}$$

Secondly, all critical cars' Power, SOC, and cumulated power at time t are computed.

$$\begin{cases} \mathrm{SOC}_i(t) = \mathrm{SOC}_i(t-1) + \dfrac{\lambda_c * \Delta t}{C_b}, \\ \mathrm{Power}(t)_{i,\mathrm{critical}} = -\dfrac{\lambda_c \times \alpha}{1000}, \\ \mathrm{Power}(t)_{\mathrm{critical}} = \sum_{i=0} \mathrm{Power}(t)_{i,\mathrm{critical}} \end{cases} \quad \forall i \in \text{Set_cars_critical} \tag{3.19}$$

To proceed, we must reevaluate W_+ and W_-. There is a possibility that too many critical cars need to be charged more than the power available, which will force other cars to discharge at time t to provide the power. The computation is as follows:

$$\begin{cases} D_+ \to D_+ & \text{if } \mathrm{P}_{\mathrm{diff}}(t) + \mathrm{Power}(t)_{\mathrm{critical}} > 0 \\ W_+ \to D_- & \text{if } \mathrm{P}_{\mathrm{diff}}(t) + \mathrm{Power}(t)_{\mathrm{critical}} < 0 \\ D_- \to D_- & \text{always} \end{cases} \tag{3.20}$$

Thirdly from the parked cars, the power and SOC of all blocked cars dependent on W_- or W_+ is computed:

$$\begin{cases} \mathrm{SOC}_i(t) = \mathrm{SOC}_i(t-1) \\ \mathrm{Power}(t)_{i,\mathrm{blocked}} = 0 \end{cases} \quad \forall i \in \text{Set_cars_blocked}, t \in D_-, D_+ \tag{3.21}$$

Sort the SOC of each car in Set_cars_available according to :

- D_-: cars are sorted in descending order by SOC.
- D_+: cars are sorted in ascending order by SOC.

Compute the Power and SOC of all available and non-critical cars at time t.

$$\begin{aligned} SOC_i(t) &= \begin{cases} SOC_i(t-1) + \dfrac{\lambda_c * \Delta t}{C_b}, & \text{if } D_+ \\ SOC_i(t-1) - \dfrac{\lambda_d * \Delta t}{C_b}, & \text{if } D_- \end{cases} \\ Power_i(t) &= \begin{cases} -\dfrac{\lambda_c \times \alpha}{1000}, & \text{if } D_+ \\ \dfrac{\lambda_d \times \alpha}{1000}, & \text{if } D_- \end{cases} \end{aligned} \quad , \quad \forall i \in \text{Set_cars_available} \tag{3.22}$$

Compute $\mathrm{P}_{\mathrm{difference}}(t)$, which is the difference between wind power generation and New York City power demand at time t.
Compute the number $\mathrm{Nb}(t)_{\mathrm{d/c}}$ of cars :

$$N_{\mathrm{d/c}}(t) = \begin{cases} \min\left\{N : \left|\sum_{i=0}^{N} \mathrm{Power}_i(t)\right| \geq \mathrm{P}_{\mathrm{diff}}(t) + \mathrm{Power}(t)_{\mathrm{critical}}\right\} & \text{if } D_+ \\ \max\left\{N : \sum_{i=0}^{N} \mathrm{Power}_i(t) \leq |\mathrm{P}_{\mathrm{diff}}(t) + \mathrm{Power}(t)_{\mathrm{critical}}|\right\} & \text{if } D_- \end{cases} \tag{3.23}$$

$$\forall i \in \text{Set_cars_available}$$

$N(t)_{\mathrm{d/c}}$ based on the constraint imposed by the sign of $\mathrm{P}_{\mathrm{difference}}(t)$. $N(t)_{\mathrm{d/c}}$ represents the desired value of N that satisfies certain conditions. We can optimize our calculations and adjust the resulting value of $N(t)_{\mathrm{d/c}}$ by incorporating the constraint of the charging or discharging state denoted by $D+$ or $D-$ and the cumulative power calculation order determined by the previous time instance's state of charge (SOC) at $t-1$. To clarify, we define two action sets: *Set_charging* and *Set_discharging*. These sets contain all the cars actively charging or discharging at a particular time t. $Set_charging(t)$ is defined as the set of cars that are available for charging and have an index less than or equal to $N_{d/c}(t)$ for D_+. Similarly, $Set_discharging(t)$ is the set of cars that are available for discharging and have an index less than or equal to $N_{d/c}(t)$ for D_-.

$$Set_charging(t) = \left\{ i : i \in Set_cars_available(t) \text{ and } i \leq N_{d/c}(t) \text{ for } D_+ \right\}$$
$$Set_discharging(t) = \left\{ i : i \in Set_cars_available(t) \text{ and } i \leq N_{d/c}(t) \text{ for } D_- \right\}$$

For all other cars denoted by i, in the set *Set_cars_available* we update the SOC and the power as follows :

$$\forall i \in \text{Set_cars_available}(t), \quad N(t)_{\mathrm{d/c}} + 1 \leq i$$
$$\mathrm{SOC}_i(t) = \mathrm{SOC}_i(t-1)$$
$$\mathrm{Power}_i(t) = 0$$

3.2.4 SCA: Total Power from V2G and Total Cars Participating

The total power coming from electric vehicles to balance out the power grid and have $\mathrm{P}_{\mathrm{production}}(t) = \mathrm{P}_{\mathrm{demand}}(t)$ is given by :

$$\mathrm{P}_{\mathrm{V2G}}(t) = \sum_{i=0}^{N(t)_{\mathrm{d/c}}} Power_i(t) + \mathrm{Power}(t)_{\mathrm{critical}} \quad \forall i \in \text{Set_cars_available} \tag{3.24}$$

Total Participation (excluding Driving and Blocked Cars)

The total amount of cars discharging/charging at a given time is given by :

$$\mathrm{Total}(t)_{\mathrm{d/c}} = (N(t)_{\mathrm{d/c}} + N(t)_{\mathrm{critical}}) * \alpha \tag{3.25}$$

This formulation clarifies that $\mathrm{Total}(t)_{\mathrm{d/c}}$ accounts for the scaled number of cars necessary to adhere to the power constraints, whereas $N(t)_{\mathrm{d/c}} + N(t)_{\mathrm{critical}}$ denotes the original count of cars selected from Set_cars. The intuitive understanding is that the scaling factor implies that each car is treated as if replicated x times, ensuring a consistent calculation basis.

Optimization: Total Participation Including Driving and Blocked Cars

From the computation in eq.(3.25), we have computed the total amount of cars needed for charging/discharging at t, but we have not considered the cars driving and blocked at time instance t.
The objective of the optimization is to minimize the total number of cars required for Vehicle-to-Grid (V2G) purposes. Therefore, a straightforward approach of multiplying the counts for driving and blocked cars by the scaling factor is insufficient. Instead, the necessary number of driving and blocked cars is determined by considering the probability of cars being in a driving

or blocked state at a time 't.'
Let N_{driving} represent the size of *Set_cars_driving*, N_{blocked} represent the size of *Set_cars_blocked* and N_{cars} represent the size of *Set_cars*. Here, it is important to remember that N_{blocked} is dependent of $\text{SOC}(t-1)_i$ as described in eq.(3.7) while N_{driving} the mobility patterns predefined and not determined by the SCA algorithm.
Let $p(t)_{\text{driving}}$ represent the probability of a car driving at t and $p(t)_{\text{blocked}}$ represent the probability of a car being blocked at t.

$$\begin{aligned} p(t)_{\text{driving}} &= \frac{N_{\text{driving}}(t)}{N_{\text{cars}}} \\ p(t)_{\text{blocked}} &= \frac{N_{\text{blocked}}(t)}{N_{\text{cars}}} \end{aligned} \tag{3.26}$$

The total number of cars required at a time 't' can be determined by the following expression:

$$\begin{aligned} \text{Total}(t) &= \frac{\text{Total}(t)_{\text{d/c}}}{1 - p(t)_{\text{driving}} - p(t)_{\text{blocked}}} \\ &= \frac{(N(t)_{\text{d/c}} + N(t)_{\text{critical}})}{1 - p(t)_{\text{driving}} - p(t)_{\text{blocked}}} * \alpha \end{aligned} \tag{3.27}$$

By dividing $\text{Total}(t)_{\text{d/c}}$ by the complement of the sum of the driving and blocked probabilities, we obtain the adjusted total amount of cars needed at a time 't.' This adjustment considers the probabilities and ensures an accurate estimation of the number of cars required for V2G operations.
To assess the potential of Vehicle-to-Grid (V2G) in New York, it is crucial to determine the maximum number of cars required over time. This metric provides valuable insights into the scale and feasibility of V2G operations. The maximal total number of cars can be calculated using the following formula:

$$\text{MaxCars} = \max_t \left(\text{Total}(t)\right) \tag{3.28}$$

Additionally, to identify the specific time instance at which the maximum total cars are required, we can compute the corresponding time value $t_{\max}$:

$$t_{\max} = \arg\max_t \left(\text{Total}(t)\right) \tag{3.29}$$

We can obtain the adjusted number of cars required by subtracting the sum of the driving and blocked probabilities from the complement.

$$N(t,\alpha)_{\text{cars}} = (N(t)_{\text{d/c}} + N(t)_{\text{critical}}) + (p(t)_{\text{driving}} + p(t)_{\text{blocked}}) * \frac{\text{Total}(t)}{\alpha} \tag{3.30}$$

The maximum value of $N(t)_{\text{cars}}$ measurement will determine the minimal size of the EV fleet in consideration: $\max_t(N_{\text{cars}})$

3.3 System Dynamics and Optimization Processes

3.3.1 System Dynamics

Following our computations and identifying relevant variables in the previous sections, examining how these variables interact is important. This section is dedicated to exploring the intricacies of these interdependencies, with a special focus on the State of Charge (SOC). A

comprehensive understanding of SOC nuances is vital as a pivotal variable that determines the current state and future actions of an Electric Vehicle (EV).

The intricacies come from the fact that the level of charge for one car (SOC_i(t+1)) is connected to the charge levels of all other cars within the system at the same time (t+1). Additionally, these connections carry over to the progression of the charge levels over time. As a result, a thorough examination requires considering the passage of time and the number of vehicles, their individual states, and actions during each specific time interval.
The function **F** formally represents the dependencies:

$$SOC_i(t+1) = \mathbf{F}(t+1, SOC_i(t), P_{diffleft}(t+1), N_{d/c}(t+1)) \quad \forall i \in Set_cars(t) \tag{3.31}$$

where $P_{diffleft}(t+1)$ is defined by:

$$P_{diffleft}(t+1) = P_{diff}(t+1) + Power(t+1, N_{crit}(t))_{critical} \tag{3.32}$$

and $Power(t+1, (N_{crit}(t))_{critical}$ defined by eq.(3.19) and $P_{diff}(t+1)$ is the power balance defined by eq.(3.1).
N_{crit} is the sum of all cars with a critical state of charge $SOC_{crit}(t)$ in Set_cars spanning from i to N_{cars}. $N_{crit}(t)$ is defined as such :

$$N_{crit}(t) = \sum_{i=1}^{N_{cars}} \mathbb{1}(SOC_i(t) = SOC_{crit}(t)) \tag{3.33}$$

The decision to charge or discharge is thus represented by :

$$D_{\text{charge/discharge}}(t+1) = \begin{cases} D_+ & \text{if } P_{diffleft}(t+1) > 0 \\ D_- & \text{if } P_{diffleft}(t+1) < 0 \end{cases} \tag{3.34}$$

as defined by eq.(4.16) where D_+ represents charging and D_- accounts for discharging.
The variable $N_{d/c}(t+1)$ defined in equation (3.23) plays a significant role in equation (3.31). It is dependent on the decision to charge or discharge, $D_{\text{charge/discharge}}(t+1)$. $N_{d/c}(t+1)$ represents the number of cars that must be charged or discharged at time $t+1$.

$N_{d/c}(t+1)$ is crucial because it represents the interconnectedness between all vehicles in the fleet during time $t+1$. When combined with the restrictions set by $Set_cars_blocked$, $Set_cars_critical$, and $Set_cars_driving$, $N_{d/c}(t+1)$ significantly affects the evaluation of the charge status.

Temporal Analysis

- $D_{\text{charge/discharge}}(t)$:
  - $D_{\text{charge/discharge}}(t)$ converges to D_- for increasing $N_{crit}(t)$.
  - $D_{\text{charge/discharge}}(t)$ adapts the sign of $P_{diff}(t)$ for decreasing $N_{crit}(t)$.
- $Power(t, (N_{crit}(t))_{critical}$:
  - $|Power(t, (N_{crit}(t))_{critical}|$ increases with increasing $N_{crit}(t)$
- $N_{d/c}(t)$:
  - $N_{d/c}(t)$ has a minimum at $P_{diffleft}(t) = 0$.
  - $N_{d/c}(t)$ increases with increasing $|P_{diffleft}(t)|$

- $|P_{diffleft}(t)|$
  - $|P_{diffleft}(t)|$ strictly increases for W_-
  - $|P_{diffleft}(t)|$ increases for $|Power(t, N_{crit}(t)| \geq P_{diff}(t)$ for W_+
  - $|P_{diffleft}(t)|$ decreases for $|Power(t, N_{crit}(t))| \leq P_{diff}(t)$ for W_+
- $N_{crit}(t)$:
  - $N_{crit}(t)$ increases with decreasing $SOC_i(t)$ with $i \in Set_Cars$, .
- Individual SOC $SOC_i(t)$ at each time step:
  - $SOC_i(t)$ for $i \in Set_charging(t)$ increases with increasing $N_{d/c}(t)$ for D_+, .
  - $SOC_i(t)$ for $i \in Set_discharging(t)$ decreases with increasing $N_{d/c}(t)$ for D_-.
  - $SOC_i(t)$ for $i \in (Set_cars_available(t) \setminus Set_charging(t))$ stays constant for D_+ .
  - $SOC_i(t)$ for $i \in (Set_cars_available(t) \setminus Set_discharging(t))$ stays constant for D_-
  - $SOC_i(t)$ decreases if $i \in Set_cars_driving(t)$.
  - $SOC_i(t)$ stays constant if $i \in Set_cars_critical(t) \cup Set_cars_blocked(t)$.
- General fleet behavior of $SOC(t)$ over time :
  - $SOC(t)$ increases with increasing $N_{d/c}(t)$ for D_+, .
  - $SOC(t)$ decreases with increasing $N_{d/c}(t)$ for D_-.
  - $SOC(t)$ decreases if $p_{driving(t)}$ increases.

The variable $p_{driving(t)}$ represents a consistent probability distribution based on mobility patterns. It varies between high and low values and typically does not have a significant effect on the overall battery life of the fleet in the long run.
Although the system's evolution may not be entirely clear, some distinct observations can be made.
As time passes, the entire system will be controlled by the duration of $W+$ and $W-$ periods and the magnitude of N_{crit}, which triggers periods of (D_-). During prolonged periods of low wind, $|P_{diffleft}(t)|$ may accumulate due to more cars entering a critical state, putting additional strain on the power grid. This requires more cars to discharge, resulting in a general decrease in the fleet's SOC. Consequently, the dynamics of the SOC over time become more important than the power constraints determined by wind and NYC demand, leading to a feedback loop with the rise of N_{crit}. Thus, the decision to discharge or charge cars is increasingly influenced by the behavior of other cars rather than the actual power generation and demand imbalance. This cycle may require a significantly larger number of cars participating in the V2G scheme to maintain stability. However, during extended $W+$ periods, driving periods regulate the saturation of excessively high SOC. Therefore, cars will always be available for charging.

In conclusion, many cars will likely be required to participate in V2G to prevent SOC levels from dropping too low. The occurence of $W+$ and $W-$ periods will likely not be enough to manage the SOC effectively. It may be necessary to introduce an additional control parameter to maintain SOC stability, which could improve the system's overall stability with a realistic EV fleet size.

3.3.2 Computational Performance Evaluation and Parameter Optimization

Our previous discussion examined the system's dynamics without accounting for parameters designed to decrease computational complexity: α and N_{cars}. Here, α represents a scaling factor that assumes the presence of α times car 'i' within the set of cars known as set_cars, with a fleet size of N_{cars}. These vehicles exhibit similar charging/discharging behavior and mobility patterns. N_{cars}, on the other hand, represents the initial fleet size. The scaling factor presents a critical parameter in our model's optimization process. As suggested by equations eq.(3.19), eq.(3.22), eq.(3.23), eq.(3.24), etc., the scaling factor exhibits a direct influence on the Total(t) and $\mathrm{P_{V2G}}(t)$ variable. Our ultimate goal is to find optimal parameters of alpha and N_{cars} that reduce computational complexity and minimize Total(t, α).

$$\begin{cases} \mathrm{Total}(t,\alpha) = \dfrac{(N(t,\alpha)_{\mathrm{d/c}} + N(t,\alpha)_{\mathrm{critical}})}{1 - p(t)_{\mathrm{driving}} - p(t,\alpha)_{\mathrm{blocked}}} * \alpha \\ \mathrm{P_{V2G}}(t,\alpha) = \displaystyle\sum_{i=0}^{N(t,\alpha)_{\mathrm{d/c}}} Power_i(t,\alpha) + \mathrm{Power}(t,\alpha)_{\mathrm{critical}} \end{cases} \tag{3.35}$$

The link demonstrated by the equations in eq.(3.35) indicates that careful adjustments to the scaling factor are necessary to balance power and minimize the number of cars. In the following discussion, we will explore these dependencies to gain a better understanding of how the scaling factor impacts the optimization process.

To ensure the stability of the system, we can see that $1 - p(t)_{\mathrm{driving}} - p(t,\alpha)_{\mathrm{blocked}} > 0$, meaning that for all time:

$$(N_{driving}(t) + N_{blocked}(t,\alpha)) < N_{cars} \tag{3.36}$$

This means that the sum of driving and blocked cars at any time t should always be within the initial dataset size of Set_cars. Based on the equations we have computed, it is clear that there are additional relationships and limitations for α that should be considered.

P0. $\mathbf{P}(t,\alpha)$ from eq.(3.2) has an $\alpha_{\min}$ for which $\mathbf{P}(t,\alpha_{\min}) > 0$ and for all t.

P1. $\mathbf{P}(t,\alpha)$ from eq.(3.2) increases with higher α. However, $\mathbf{P}(t,\alpha)$ should stay small to satisfy the grid's power balance.

P2. $N(t,\alpha)_{\mathrm{d/c}}$ and $N(t)_{\mathrm{cars}}$ increase with smaller α.

P3. Total(t,α) converges to a value with decreasing α and increasing N_{cars}.

P4. Reduction of computational complexity: small N_{cars} and high α.

We first address the properties [P0 to prioritize the constraints.], [P1.]. Our objective is to maintain the power balance of the grid, which involves minimizing $\mathbf{P}(t,\alpha)$. A smaller scaling factor provides finer control, as each car can contribute only a limited amount of power. [P0.] gives a hard constraint: α can not be chosen smaller than $\alpha_{\min}$. [P1.] shows that we also have a limitation on $\alpha_{\max}$ to ensure minimal values of $\mathbf{P}(t,\alpha)$. Formally, this scenario can be captured through a constraint optimization problem, which can be expressed mathematically as follows:

$$\begin{aligned} \underset{\alpha}{\text{minimize}} \quad & \mathbf{P}(t,\alpha) \\ \text{subject to} \quad & \alpha_{\min} \le \alpha \le \alpha_{\max}, \\ & \mathbf{P}(t,\alpha_{\min}) > 0, \quad \forall t. \end{aligned} \tag{3.37}$$

Furthermore, we aim to minimize the number of cars required for V2G operations. We have identified a key pattern illustrated in property [P3.], which shows that there is a specific value

α_{conv} where the function $\text{Total}(t, \alpha_{\text{conv}})$ starts to converge. Although the model is inherently stochastic, the trend towards a constant value is significant even if the function does not converge precisely. Mathematically, this is represented as:

$$\lim_{\alpha \to \alpha_{\text{conv}}} \frac{d, \text{Total}(t, \alpha)}{d, \alpha} \approx 0 \tag{3.38}$$

Moreover, according to property [P2.], a smaller scaling factor requires a more extensive set of unique EV trajectories, denoted as N_{total}. Although reducing α and increasing the dataset size improves the model's accuracy, it also significantly augments the computational complexity and execution time of the SCA, as highlighted in property [P4.]. To sum up, this section has explained how several factors interact to maintain the power balance in our grid system. We have highlighted the importance of the scaling factor, denoted by α, which needs to be within certain limits to avoid instability in the system. Additionally, we have explored the consequences of aiming for more precise modeling and finer control by reducing α and increasing the dataset size N_{cars}. While these choices make the system more realistic, they also significantly increase the computational workload. In the next section, we'll create a mathematical equation to represent our optimization problem. This approach will help us balance various elements and improve the efficiency of Vehicle-to-Grid operations.

3.3.3 Formulation of the Smart-Charging Optimization Problem

Firstly, we aim to find the best scaling factor, which we call α_{opt}. This factor should keep the power balance of the grid and allow the Smart Charging Algorithm (SCA) to work efficiently. To achieve this, we need to minimize the power function $\mathbf{P}(t, \alpha)$ while following the rules stated in [P0.]-[P4.]. Mathematically, we can express this objective as follows:

$$\underset{\alpha}{\text{minimize}} \quad \mathbf{P}(t, \alpha) \quad \text{subject to} \quad \alpha_{\min} \leq \alpha_{\text{opt}} \leq \alpha_{\text{conv}}, \quad \mathbf{P}(t, \alpha_{\min}) > 0, \quad \forall t. \tag{3.39}$$

The value of α_{conv} is calculated using equation (3.38). On the other hand, $\alpha_{\min}$ is determined by the condition that $\mathbf{P}(t, \alpha_{\min}) > 0$, which is necessary for maintaining power balance at all times.

Our study's secondary goal is to evaluate how many unique cars are needed at $t_{\max}$, which is represented as $N(t_{\max}, \alpha_{\text{opt}})_{\text{cars}} = N_{\text{cars}}(\alpha)$. Here we need always to ensure the stability of the SCA algorithm defined by eq.(3.36). Next, we want to determine the minimum number of vehicles required to achieve our power balance goals while reducing the need for enhanced computational power. Mathematically, this problem can be formulated as follows:

$$\begin{aligned} \underset{N(t,\alpha)_{\text{cars}}}{\text{minimize}} \quad & N(t, \alpha)_{\text{cars}} \\ \text{subject to} \quad & N(t_{\max}, \alpha_{\min})_{\text{cars}} \leq N(t_{\max}, \alpha_{\text{opt}})_{\text{cars}}, \\ & N(t_{\max}, \alpha_{\text{opt}})_{\text{cars}} \leq N(t_{\max}, \alpha_{\text{conv}})_{\text{cars}}. \end{aligned} \tag{3.40}$$

This equation calculates the total count of unique cars at $t_{\max}$ with a scaling factor of α_{opt}, represented as $N(t_{\max}, \alpha_{\text{opt}})_{\text{cars}}$. The requirement is that this value should fall within the range of $N(t_{\max}, \alpha_{\min})_{\text{cars}}$ and $N(t_{\max}, \alpha_{\text{conv}})_{\text{cars}}$.

Hence, our primary goal is to find the best combination of α_{opt} and $N(t_{\max}, \alpha_{\text{opt}})_{\text{cars}}$ to decrease the SCA's computational complexity while ensuring stability at all times. At the same time, we want to ensure the accuracy of controlling the power balance with our SCA.

3.4 Results and Discussion

All codes can be found on Git Hub and are followed by a Read-me file to reproduce the files and plots introduced in this study.

3.4.1 Variable Definitions and Parameter Evaluation

In the following section, the performance of our Smart Charging Algorithm (SCA) will be delineated, focusing on determining the requisite number of vehicles for the Vehicle-to-Grid (V2G) system in New York to be feasible. We will do a parameter sensitivity analysis for all following scenarios and vary values from the scaling factor: α, discharging rate: λ_d, charging rate: λ_c, and wind power output capacity: C_W.

Fixed Variables

For all scenarios under consideration, the subsequent three parameters will remain invariant:

- Time interval (hours): $\Delta t = 0.25$
- Energy loss from driving (kWh): $\Delta D = 3$
- Battery capacity (kWh): $C_b = 70$

We discovered a database of all-electric vehicles in Europe, presented in [34]. This database provides valuable information about battery capacity, energy consumption, and the availability of bidirectional charging. The variable ΔD is based on an average energy consumption of 0.2 to 0.35 kWh per mile, as reported in [34]. Additionally, the speed limit on major roads in New York City is typically between 25 and 30 miles per hour (40 and 48 km per hour). Still, the average speed can be much lower during rush hours and in areas with heavy traffic, ranging from 5 to 15 miles per hour (8 and 24 km per hour) [35]. The patterns described in Chapter 2 indicate high mobility during peak hours, so we chose an average speed within the lower range. Based on information from [34] and [35], we define the following:

$$\begin{aligned} \text{Energy consumption(kwh/mile):0.3} \\ \text{Average speed (miles/hour):10} \\ \text{Energy loss from driving (kWh)=Average speed*Energy consumption=3} \end{aligned} \tag{3.41}$$

The battery capacity denoted as C_b, is determined by the average battery capacity of existing electric vehicles (EVs). According to the data set from [34], the average battery capacity is around 70 kWh. The reason for keeping these variables constant is that they are relatively independent.
Although these factors may impact the model's results, their effects follow a predictable pattern, reducing their significance for sensitivity analysis. For example, an increase in battery capacity minimizes the need for cars, while an increase in energy loss during driving increases the demand for vehicles. The choice of the time interval is based on computational considerations, where smaller intervals provide more accurate results but also increase computational complexity.

Charging and Discharging Rates: λ_c, λ_d

As highlighted in the introduction, we encounter various levels of charging capabilities. A list of all major bidirectional chargers in the US with their respective maximum power rates is provided in [14].
By observing various EVs that support bidirectional charging, we can estimate a realistic power range for discharging is between 7kw-34kw [14]. Given the future development plans of charging stations, we run our model with a DC Fast charger Level 3, where we choose the lowest

charging rate for Level 3 DC charging of 50 kW and a discharging rate of 20 kW. Choosing a lower discharging rate comes from the technical limitations of bi-directional charging. Also, it will account for a delay in our model, considering the "range anxiety" of EV owners [36].
The technical specifics of the charging and discharging rates might need reconsideration depending on the technical limitations of the grid and infrastructure in New York City. They can easily be adjusted in the algorithm.

Wind Capacity Factors: C_W

In Chapter 2, we presented a modeling methodology for offshore wind power generation that anticipates a peak capacity of 9000MW [4]. Our model aims to explore the potential changes in the requisite number of vehicles participating in V2G with the manipulation of the power capacity of offshore wind farms. We have analyzed the power capacity of offshore wind farms and found that the duration of low wind and high wind periods is more critical than their occurrence. Changing the capacity factor may not directly impact as it would only shift the power output and decrease the frequency of low wind periods but not the duration of these periods.

Concluding we will only inspect two different power capacity factors :

- 13000 MW
- 9000 MW

3.4.2 Results and Plots

In each scenario, we recompute the optimal α from the previous chapter as per eq.(3.40). We perceive 9000 MW as the most realistic wind capacity. First, we assess the impact of the charging/discharging rates. This leads to the following scenarios:

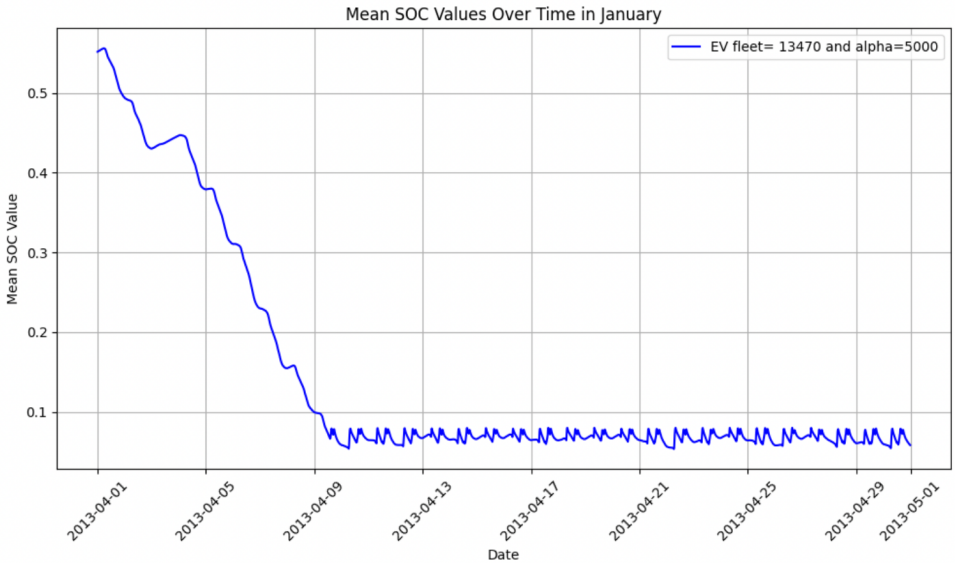
- Scenario 1.1: 9000 MW with 50Kw charging and 20 Kw discharging in April
- Scenario 1.2: 9000 MW with 50Kw charging and 20 Kw discharging in January

By analyzing these scenarios, we expect to understand the dynamics influencing the total number of cars needed. We run the SCA algorithm for each month independently and determine alpha from eq.(3.40) for each month. It's important to mention that we also ran both scenarios with a capacity factor of 13000 MW. However, we didn't notice any significant changes in the results, so we won't explicitly display them.

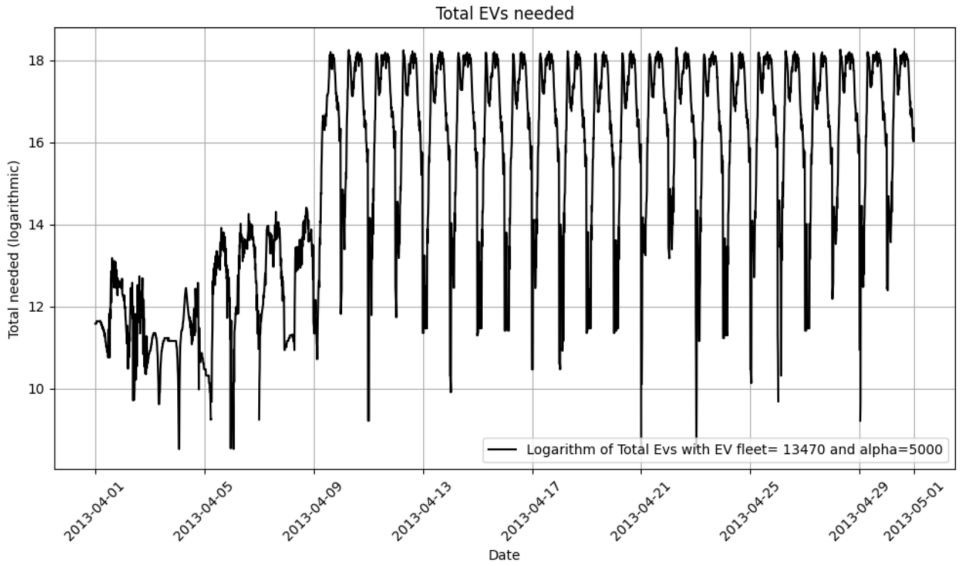
Results

Our initial Smart-charging (SCA) algorithm demonstrated disappointing outcomes, largely due to our inability to fine-tune the scaling values for alpha and the varied number of vehicles. Our approach for fine-tuning is described in this section.
As we seek to identify the optimal choice of α and N_{cars}, we produce a variety of datasets of our mobility and SOC_{min} with varying sizes. The method for scaling our mobility dataset is presented in Chapter 2. We run our algorithm with different α and different N_{cars}. For computational feasibility, we start with high α and low N_{cars} values and check for robustness by ensuring that eq.(3.36) is satisfied for all t. If stability is not ensured, we slowly increase N_{cars} and keep α constant until we have exhausted all computational power. We established that a dataset of the size $N_{cars} = 26940$ already leads to an exhaustion of computational power. We will look for increasing alpha that ensures this stability. After evaluating various combinations of parameters within the range of computational feasibility for N_{cars}, we have not found a suitable combination of α and N_{cars} for which our system remains stable. We will show that this

is due to the SOC of the fleet of cars i, converging to 0 over time as seen in Fig.(3.1)(a).
In Fig.(3.1)(a), we have depicted the mean SOC value overall EVs in a dataset of size $N_{cars} = 13470$ and $\alpha = 5000$ over April with an offshore wind farm of capacity=9000 MW. In April, as previously derived in the table in Fig.(2.3), we have an offset of +227.27 Gwh. Once the SOC hits values nearing 0 or 1, it's classified as 'blocked' by the defined condition if it hasn't already been identified as in a critical state. As a result, there comes a point when most cars from our initial dataset are either categorized as 'blocked' or as critical. This can be seen in Fig.(3.1) at the time 2013-04-09, where we have exhausted the SOC value to critical states. As the number of critical cars increases, the requirement for more available cars for charging/discharging to maintain power balance intensifies, creating a vicious circle.
This can mathematically be described by the stability equation eq.(3.36), where $N_{driving}(t) + N_{blocked}(t)$ nears N_{cars} and hence for a time $t = t_{false}$ when $N_{driving}(t) + N_{blocked}(t) > N_{cars}$ the algorithm's functionality falters. For April, with a given EV fleet size of $N_{cars} = 13470$ and $\alpha = 5000$, this can be observed at $t =$2013-04-09.
$N_{driving}(t)$ solely depends on mobility behavior, whereas $N_{blocked}(t)$ relies on the SCA algorithm and tends to increase during extended periods of charging/discharging. Therefore, it's contingent on the unalterable dynamics of wind power generation. The only room for modification lies in selecting N_{cars}, representing the size of the initial dataset (number of cars), which must be sufficiently large to ensure stability.



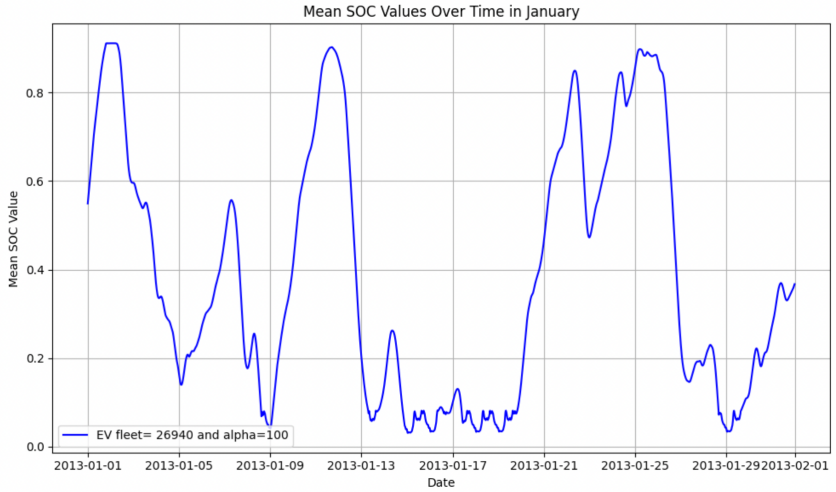
(a) Convergence of the mean SOC overall EVs in the fleet with $N_{cars} = 13470$ and $\alpha = 5000$



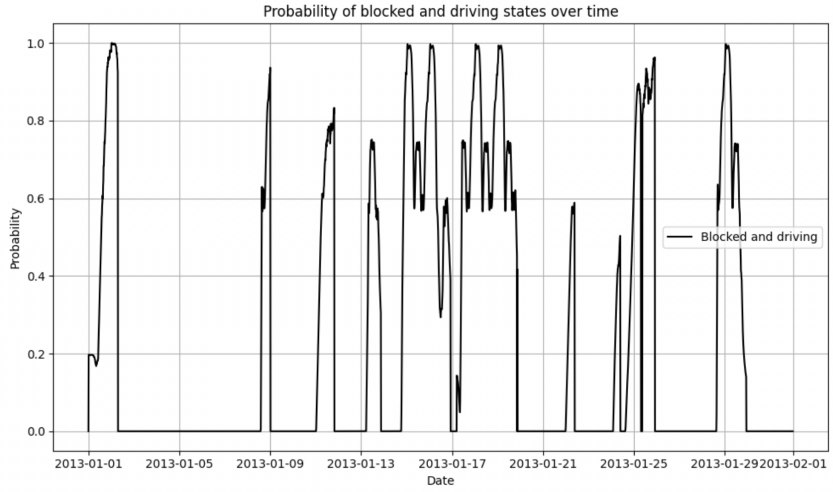
(b) Total number of EVs needed in fleet with $N_{cars} = 13470$ and $\alpha = 5000$

Figure 3.1: Proof of instability of SCA algorithm with converging SOC and overshooting total number of EVs needed

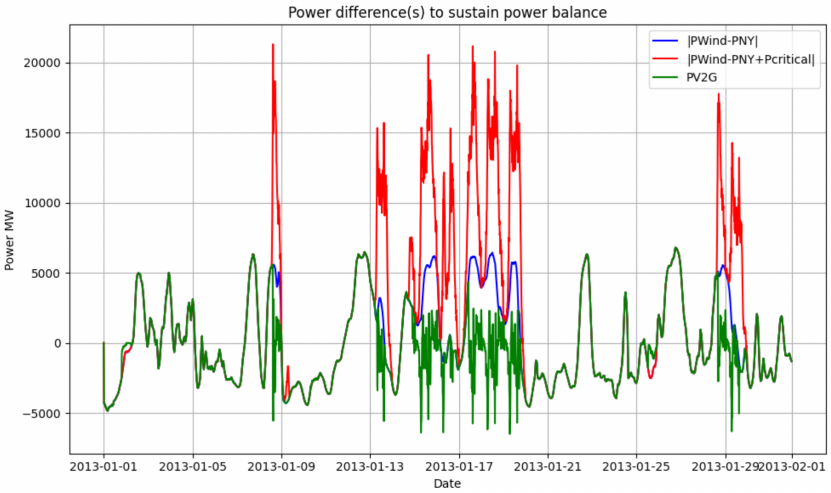
We extend our analysis to higher N_{cars} and lower α. For N_{cars}=26940 and α= 100, we also observe instability running our SCA algorithm over January (-217.73 Gwh). For better intuition of the potential, we slightly changed our code to ensure stability instead of discharging/charging as many cars as needed to maintain the power balance without considering the available vehicles. We only discharge/charge the available cars and then compute how many are missing to keep the power balance. In this scenario, the stability equation eq.(3.36) will always be satisfied, but we will not always ensure power balance.



(a) Mean SOC overall EVs in the fleet with $N_{cars} = 26940$ and $\alpha = 100$

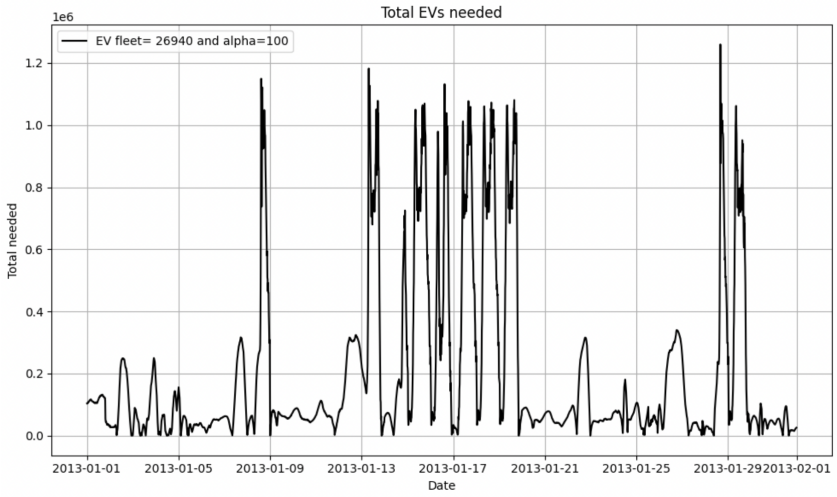


(b) Sum of driving and blocked probability of EVs ensuring stability at all times with $N_{cars} = 26940$ and $\alpha = 100$

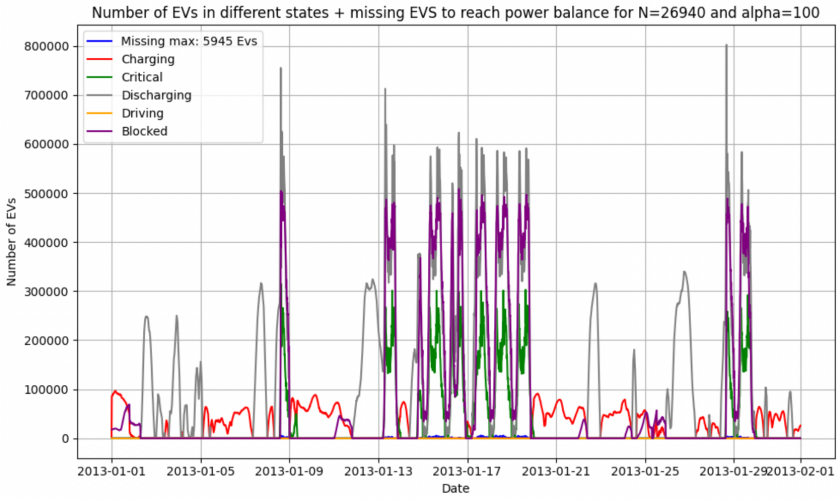


(c) Power difference and PV2G comparison with $N_{cars} = 26940$ and $\alpha = 100$

Figure 3.2: Proof of stability of SCA algorithm with non-saturating SOC but power balance not fulfilled



(a) Total number of EVs needed in fleet with $N_{cars} = 26940$ and $\alpha = 100$



(b) Number of EVs in different states over time

Figure 3.3: Total number of Evs needed and distribution of EVs into different action states

In Fig.(3.2), we have shown that our system is stable and, therefore, our results are accurate. However, we could only ensure stability by revoking the hard constraint of the power balance and instead discharging/charging as many cars as available in the dataset. We can observe that in various periods in January, we are not able to cover NYC demand (t=2013-01-09 and from t=2013-01-13 to 2013-01-20) with N_{cars}=26940 and α= 100. During these periods, it is essential to involve additional electrical vehicles to meet the required power demand. In Fig.(3.2), we can already observe that we would need at least 1.3 million EVs to cover NYC electricity demand during non-critical periods. However, considering the significant power difference shown in Fig.(3.3), we anticipate an even higher participation number will be necessary.
Concluding, finding the optimal combination of N_{cars} and α is not trivial.

Our current strategy of scaling up a fraction of available cars without considering each vehicle's state of charge (SOC) dynamics is an apparent oversight. We must involve more cars to prevent SOC overloading and maintain power equilibrium. Although our technique can work with the correct initial dataset size and alpha value, it becomes computationally complex when dealing with high vehicle demand for Vehicle-to-Grid (V2G) scenarios that exceed one million cars. To

overcome this limitation, we have developed a new approach that models SOC dynamics, EV mobility patterns, and power exchange with V2G.

Chapter 4

Reassessment of the Smart-Charging Algorithm

In the last chapter, we learned that the Smart-charging Algorithm (SCA) is a complex optimization problem due to the interdependent variables and instability. Unfortunately, due to limited computational power, it was not possible to calculate the optimal parameters for equation (3.36) to ensure SCA stability.
In this chapter, we will be sharing an algorithm that has been updated to guarantee stability for any given parameters while also maintaining power balance. We aim to minimize the computational power required to achieve this. All necessary codes are accessible on the Git Hub.

4.1 Algorithm Description and Improvements

The current algorithm is based on the same assumptions and constraints as the previous SCA algorithm but with a different implementation of the scaling factor. Instead of using a constant scaling factor α, we now compute a new α value for each time step. Computing $\alpha(t)$ value will ensure that the power balance is fulfilled at all times and that $\mathbf{P}(t, \alpha(t))$ from eq.(3.2) is minimal. We will generate the corresponding attributes from Set_cars defined in eq.(3.4) for each car scaled with alpha. This section will provide pseudocodes and a detailed description of the newly invented algorithm.
In Alg.(1), we have not altered our previous algorithm and have only established different sets of cars to proceed with the discharging and charging process. We define df_cars as the data frame holding the availability matrix of driving/parked of each car, df_SOC: as the data frame storing the SOC value of each car at time t and $t+1$, $Pdifference$: the power difference between Wind and NYC demand and df_SOC_min: the data frame storing the minimal State of Charge.

Algorithm 1: Compute SOC

Result: Compute SOC
Require $df_cars, df_SOC, df_SOC_min$;
for t *in* $df_cars[1:]$ **do**
    Compute mask for car states;
    $parked_prev \leftarrow$ previous state in df_cars_month equals 1;
    $driving_prev \leftarrow\sim parked_prev$;
    Identify critical cars;
    $critical \leftarrow$ those parked cars at $t-1$ that after potential discharging would reach SOC_min and are driving at t;
    Get indices for these critical and parked cars: $critical_parked_indices$;
    Identify blocked cars;
    $condition2 \leftarrow$ those cars that at $t-1$ are parked, not critical, and can't be fully charged in the next timestep;
    $condition \leftarrow$ those cars that at $t-1$ are parked, not critical, and have enough Charge to be discharged in the next timestep;
    Update SOC for all cars at t from driving and critical;
    For cars that were driving, reduce SOC considering energy loss due to driving;
    For critical, increase SOC considering charging;
    Get $Pdifference$ for time $t-1$;
    Compute power of all critical cars $P_critical$ at $t-1$;
    Compute remaining power difference $Pdifference_left$ by adding $P_critical$ to $Pdifference$ at $t-1$;
end

Alg.(2) distinguishes between the charging and discharging periods:

Algorithm 2: Charge or discharge EVs

Result: Charge or discharge EVs
Require $df_cars, df_SOC, df_SOC_min, df_15$;
for t *in* $df_cars[1:]$ **do**
    Compute remaining power difference after charging critical cars;
    if *remaining power is positive* **then**
        Maintain SOC of blocked cars;
        Charge cars that can be charged;
    else
        Maintain SOC of blocked cars;
        Discharge cars that can be discharged;
    end
end

The revised charging function now has the format:

Algorithm 3: Charging Function Pseudocode

Result: Charge Electric Vehicles

Function `Charge`(*df_SOC_, Pdifference_left, rate, delta_t, battery_capacity*)**:**

```
Sort df_SOC_ in ascending order;
if Pdifference_left is nearly balanced then
    Do not charge any other car;
    SOC remains constant;
    Count the number of cars left uncharged(all);
else
    Compute needed =(abs(Pdifference_left) * 1000/rate) − 1;
    if needed is greater than the length of SOC_sorted then
        Compute alpha = (needed/len(SOC_sorted)) − 1;
        Compute Nb as (needed/alpha) − 1;
        Handle special cases for alpha and Nb;
    end
    else
        Set alpha to 1;
        Set Nb to needed;
    end
    Charge the necessary number of cars;
    Compute and update the remaining power to be balanced;
    Count the number of cars left uncharged;
end
return Updated power difference, relevant car indices, alpha, Nb, updated
  PV2G_charge, number of uncharged cars;
```

At each time step, α is recalculated based on the requirement for $Pdifference_left$, resulting in a combination of α and $Nb = Ncars$. Nb is the number of cars from the dataset charging, and *needed* is the total number of cars required. Our process for discharging cars is similar to that of charging them. However, we won't provide specific discharging calculations here to avoid repetition, but the function details can be found on Git Hub. To determine the required number of cars charging/discharging and alpha, we create new car trajectories that incorporate their minimum SOC values. These trajectories and their SOC dynamics are crucial in tracking each car's involvement in the V2G system at any given time. This method allows us to generate new SOC values and car trajectories without prior knowledge of the exact number of cars needed, thus ensuring the computational stability of the system even during extended periods of low wind.

Initially, we replicate the dataset provided at a specific time point (t) of all relevant cars (charging/discharging) by an alpha factor. Each duplicate is assigned a distinct and exclusive car ID. Nonetheless, it is crucial that every new car ID can be linked to the original one. We presume we possess alpha instances of the same car with identical mobility patterns, as all the vehicles exhibit uniform mobility patterns. Still, we do not assume all replicates have the same SOC behavior at time t+1. Firstly, we will calculate the State of Charge (SOC) for each car in our initial dataset at the next time (t+1), using the standard expression. Furthermore, we will introduce a new factor: newly generated cars will be assigned a random SOC at t+1. The SOC values will be between 0.2-0.5 when the wind is high (W+) and 0.7-1 when the wind is low (W-). This is because these new cars need to be available (not blocked or critical) to participate in the charging and discharging processes.

Secondly, it's crucial to remember that any car might reach a critical state at any time. This means that we need to account for the future mobility behavior of each new replicated car

and, more importantly, its minimum SOC at time t and t+1, which will ensure that this car can complete its future journeys. Since we assume each new (replicated) car shares the same mobility patterns as the original car it's copied from, we can assign the corresponding mobility patterns at time t and t+1 to each new car ID. Similarly, we can access its pre-calculated minimum SOC. As each car id is traceable to its original car id, these attributes can be easily found for each new car id. We use a mapping dictionary to keep track of new and replicated car IDs. This ensures that both old and new IDs always have the necessary attributes. These attributes include minimum SOC at time t and t+1, mobility behavior at time t+1 and t, and the SOC state at time t. This allows us to determine the State of each car at any given time. The procedure is described by Alg. (4).

Algorithm 4: Update Datasets considering alpha scaling Pseudocode

Result: Update Datasets considering alpha scaling

Function `updateDataWithAlpha`(*df_cars, time, alpha, powerDiff, remainingPowerDiff, condition, condition2, critical, previousParked, previousDriving, numCars*)**:**

```
Initialize tracking for original and new car IDs and non-relevant columns;
if car replication needed (alpha > 1) then
    Generate a new row for SOC(t+1) for replicated cars with new car id's;
    Combine SOC at t+1 for all cars (both original and replicated);
    Replicate car data(driving/parking) and minimum SOC at t and t+1 points,
     considering alpha scaling;
    Create new IDs for replicated cars, which can be traced back to the original car IDs;
    Combine new and old car data and SOCmin data;
    Track SOC for all car IDs;
else
    no replication needed (alpha)=1;
    Add a new row in SOC at the next time point and update SOC; depending on
     charging/discharging;
    Keep track of SOC for all car IDs;
    Keep car data and minimum SOC data updated as usual;
end
Calculate total car participation needed for power balancing;
return Updated car datasets, updated SOC tracker;
```

From here, we output the number of total cars needed at each time step resulting from :

$$Total(t) = \alpha * N_{d/c}(t) + N_{blocked}(t) + N_{critical} + N_{driving} \tag{4.1}$$

our desired output would be

$$MaxTotal = \max_t (Total(t))$$

.

The dataset size being considered changes for each t, based on the car's power requirements for charging or discharging. This reduces the need for high memory compared to the previous SCA algorithm. Before, the dataset contained the entire driving history and critical states of all required EV trajectories, resulting in a more significant, fixed memory demand.

4.1.1 Computational Bottleneck

Despite our efforts to optimize the algorithm's runtime, our primary challenge is the algorithm's memory storage and runtime feasibility. We used techniques such as masks, data frames, and

parallelized computations. However, the implementation has to be more or less sequential due to the system's nature, where each parameter is subjected to various time-dependent constraints. Consequently, timesteps with larger data (resulting from increased α) pose significant computational demands.

In our algorithm, at each timestep t we output the total number of cars required and keep track of these values and the attributes of each car for $t+1$. Each car has five attributes: (SOC(1), parking/driving(2), SOCmin(2)) defined by eq.(3.4) for $t+1$. So for each timestep, we need to have memory storage up to:

$$\Sigma(t) = \text{Total}(t) \times 5 \tag{4.2}$$

where $Total(t)$ is defined by eq.(4.1).
We anticipate exhausting our memory storage capacity once the number of cars surpasses a certain threshold (e.g. 2 million), equivalent to approximately 10 million data points per timestep. Nevertheless, considering the real-world constraints of limited EV availability, the code's limitations are not overly critical. It aligns with the conclusion that it is not realistically feasible to maintain a power balance in New York City with a 9000MW Offshore Wind farm solely using the available cars in the city. As of 2023, approximately 2.4 million vehicles are operational in New York. Based on predictions, by 2050, the EV penetration rate is estimated to range between 40% and 60% [15]. This would correspond to 960,000- 1.44 million EVs in New York City by 2050.
For example, if the number of cars exceeds 1.44 million and our algorithm fails due to low memory storage, it would suggest relying on electric vehicles (EVs) to store all energy produced by wind farms and meet New York City's energy demand during periods of low energy production from offshore wind farms is not feasible.

4.2 Results

We executed our algorithm for different parameters. The most intuitive scenarios to evaluate the total EVs needed would be derived from the power balance offset. As already laid out in Chapter 2, we will therefore look closely into the following months:
Optimal scenario:

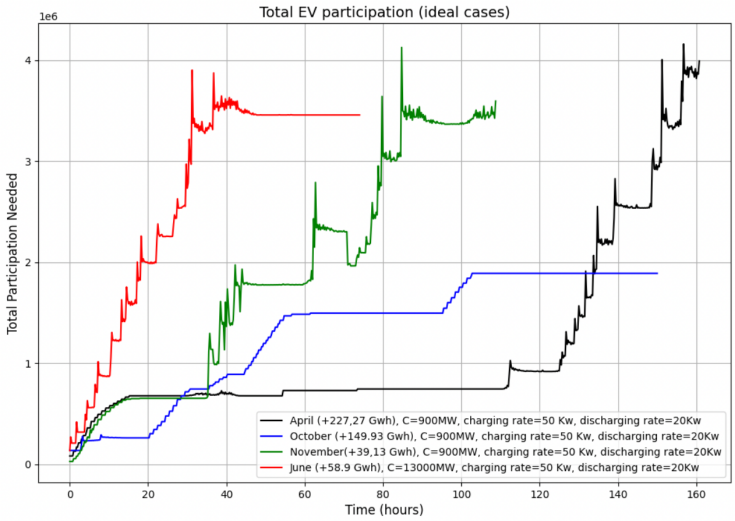
- 9000MW in April(+227.27 Gwh), October(+149.93 Gwh) and November(+39.13 Gwh)
- 13000MW in June (Offset: +58.9 Gwh)
- 9000MW in March(-22.68 Gwh), December (-100.06 Gwh)

We will also consider months with a small negative Offset to asses the feasibility :

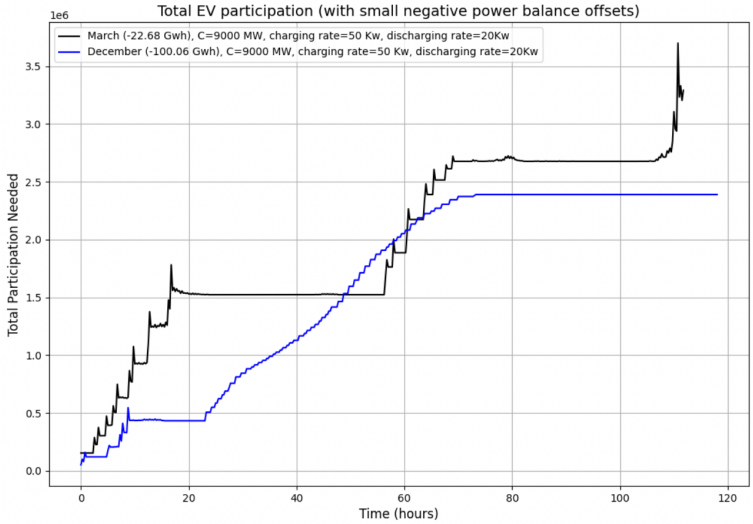
- 9000MW in March (-22.68 Gwh), December (-100.06 Gwh)

For critical cases, we would like to consider the following cases.

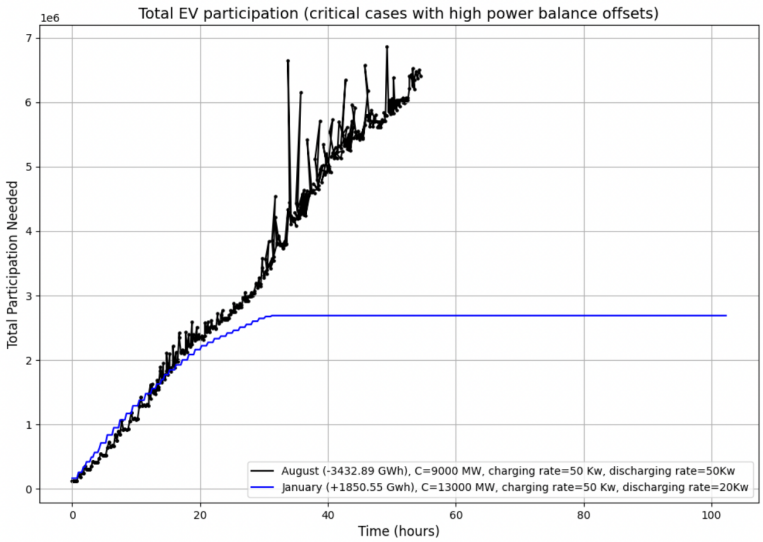
- Very low offset: 9000MW in August (-3432.89 GWh)
- Very high Offset: 13000MW in January(+1850.55 Gwh)



(a) Total EV participation for small positive power balance offset



(b) Total EV participation for small negative power balance offset



(c) Total EV participation for high power balance offset

Figure 4.1: Total EV participation

We illustrate the evolving requirements for total EV participation over time in Fig.(4.1). Fig.(4.1)(a) displays our optimal scenarios for V2G feasibility in New York City, characterized by a slight positive power offset over the months. Nevertheless, these scenarios necessitate a high volume of EV participants. For instance, in April, with the highest offset of +227.27 GWh, we observed a requirement for up to 4 million cars beyond 160 hours. In November, a period with a small positive offset, the requisite participation reached 3.5 million after 100 hours. Similarly, in June, with a capacity factor of 13000MW, we achieved 3.5 million EV participants just after 40 hours. October presented a more reasonable EV participation scenario, requiring only 1.8 million EVs beyond 140 hours. Fig.(4.1)(b) portrays the months subjected to slight negative power balance offsets. For instance, in March, with a minor negative offset of -22.68 GWh, we observed a need for up to 3.5 million cars beyond 110 hours. Despite a more substantial negative offset of 100.06 GWh in December, we observed a need for 2.5 million EVs after 120 hours. There is a significant negative power balance offset of -3432.89 GWh in August. This results in an ongoing need for more participation from electric vehicles, with the number of cars reaching 7 million after 60 hours. Lastly, for January, with a capacity of 13000MW and a substantial offset of +1850.55 GWh, we established a requirement of 2.6 million EVs after 30 hours. Based on our findings and the projected evolution of $Total(t)$ over time derived from the system dynamics analysis in Chapter 3, we anticipate encountering even higher values as time progresses. Thus, we can confidently say that, even when considering only months that overall present a positive power offset, we anticipate the need for 3-4 million cars to stabilize the power grid. This applies when integrating an offshore wind farm of 9000MW or 13000MW into New York's energy infrastructure.

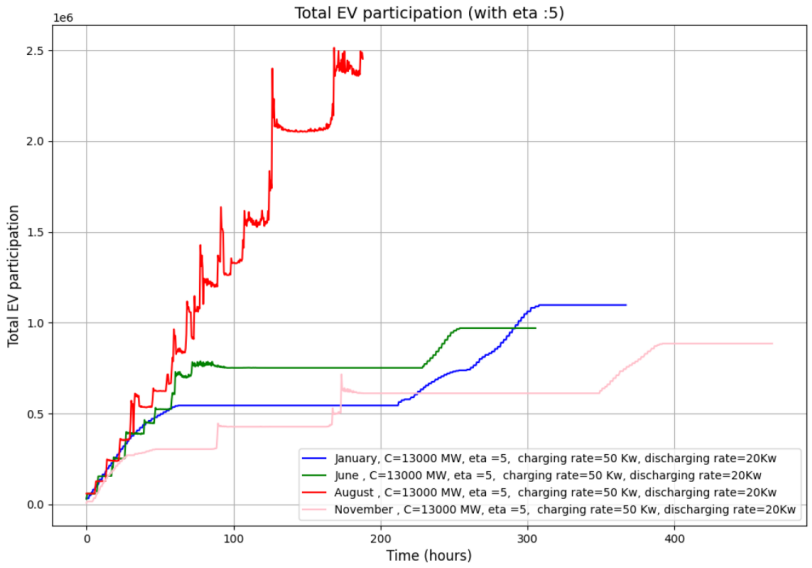
Our findings suggest that with the existing utilization of EVs, balancing New York's power demand purely through these vehicles' storage and discharging capabilities is not feasible. However, the potential for V2G technology in New York is not entirely negligible. With fewer than 2 million EVs, we could still harness V2G technology to manage at least some power surplus or deficiency situations.

For this reason, we looked into "falsely" changing the wind output curve to reduce the number of cars needed. It is essential to note that this modification would increasingly decouple the state-of-charge (SOC) and mobility of EVs in New York City from the natural wind dynamics. Instead, they become linked to the NYC demand power curve with an offset of $\frac{(+/-)P_{diff}(t)}{\eta}$. The new power curve is therefore defined as $P_{new}(t) = \frac{P_{diff}(t)}{\eta} + P_{NY}(t)$ where $P_{diff}(t) = P_{wind}(t) - P_{NY}(t)$. This would mean that with increasing η, $P_{new}(t)$ would converge to $P_{NY}(t)$, which would not be a realistic scenario for true wind speed dynamics unless the wind power generation can be controlled by other means.
Our objective was to achieve the following outcomes:

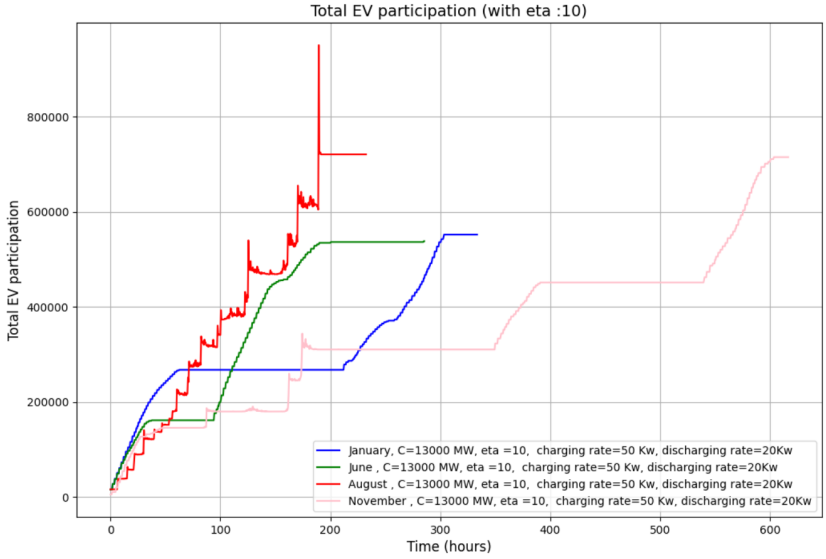
- Produce results that help develop a better understanding of the true potential of V2G
- Reduce memory storage and run time
- Identify a potential power generation curve that would make Vehicle-to-Grid (V2G) technology more feasible in New York City
- Serves as a confirmation that the algorithm was correctly implemented and lacks memory storage to compute for all t when $\eta = 1$

So to get an intuitive idea of the true potential of 1.6 million EVs in terms of helping to balance out the power grid, we would have to asses different values of η starting at low values.
We choose $\eta = 5, 10$ and evaluate for month$= 1, 6, 8, 11$ with an initial capacity factor of C=13000MW.

Analyzing the data, we noticed that even with a set value of $\eta = 5$, the number of electric vehicles (EVs) participating in the system exceeded 2.5 million within a week in August. This number reached 900,000 after two weeks in November and is expected to increase to 1-1.5 million in the remaining days of the month. Similar observations were made for June and January, where we anticipate a steeper increase and a need for 1.5-2 million EVs by month-end. Fig.(4.2)(b) shows that with $\eta = 10$, we require approximately 600,000 EVs after half of January, June, and November. By the end of the month, we can expect 800,000-1.2 million EVs. We plotted the wind power with $\eta = 5$ against NYC demand to better understand the power output dynamics in the Appendix Fig.(1). We observed that a feasible vehicle-to-grid integration in New York would require a much small power generation offset and more fluctuating wind periods.



(a) Total EV participation with $\eta = 5$



(b) Total EV participation with $\eta = 10$

Figure 4.2: Total EV participation with altered wind power output generation

The major issue with the current system is the prolonged discharging periods due to a buildup

of critical cars during low wind periods. This destabilizes the system and would require all EV fleets to be self-sustaining. The decision of an individual car would be almost entirely decoupled from actual periods of high and low wind (W- and W+) and would mainly be determined by the criticality of cars in the fleet. The results show that V2G still requires substantial EV participation even with low Power discrepancies.

4.2.1 Potential Improvements

It may require advanced problem-solving methods from computer science to tackle the computational hurdles of limited memory storage. Dynamic programming, particularly in languages such as C++, has demonstrated greater efficiency in dealing with these problems. One way to significantly reduce memory storage is by not storing SOC min and mobility/parking at t and $t+1$ for all replicated cars at every timestep. Instead, using pointers and addresses in C++ for the newly generated car ids, that can access the correct values of the original mobility dataset and SOC min to determine parked and criticality states without storing the newly generated ones. This can be quickly done in C++ with pointers and addresses to assign the correct value. However, it would require several steps in Python, such as inverting the mapping dictionary, storing relevant indices, and creating masks. This method can reduce memory storage by a scale of 5, significantly reducing the runtime.

Chapter 5

Conclusion and Outlook

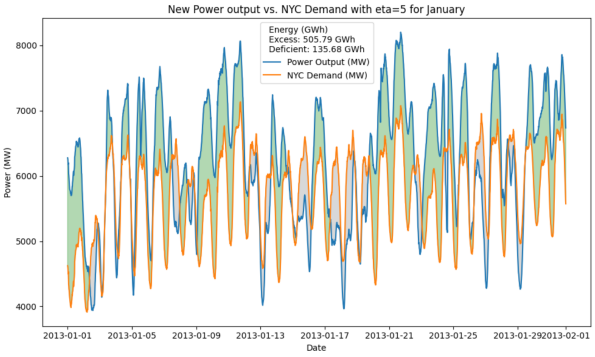
Throughout this thesis, we have studied the feasibility of incorporating Vehicle-to-Grid (V2G) services in New York City, particularly regarding the proposed 9000MW offshore wind farm in Long Island. Our research involved the development of mobility patterns that simulated vehicle parking and driving behavior for a year. Additionally, we created a year-long wind power output generation curve. We then synchronized the charging and discharging behavior of an electric vehicle (EV) with the power demand to maintain power grid equilibrium. As a result, we were able to present a V2G use scenario in New York City where an EV continuously operates in one of four states: charging, discharging, driving, or blocked. Our primary objective was to identify the number of EV participants required to sustain power equilibrium when incorporating wind-renewable energy sources. Throughout our research, we outlined various approaches, emphasizing deriving key dynamics of an EV fleet's State of Charge (SOC) involved in V2G. We also laid out the system boundaries and defined various parameters to facilitate realistic simulations. However, computational viability mostly limited our simulations, including memory storage and processing power. Nevertheless, we achieved our primary goal of evaluating the potential of V2G in New York City, as the simulations provided satisfactory results in defining the magnitude of EV participation. Unfortunately, our findings suggest that incorporating V2G and offshore wind farms to maintain the power grid's equilibrium primarily is only partially viable. It would require approximately 5-6 million electric vehicles to preserve the power balance in winter, spring, and fall, and over 7 million in the summer. We also assessed the potential of V2G with maximum participation of 1.6 million EVs and calculated the required power output curve. This indicated that a feasible integration of V2G in New York City would only be possible with a power curve that exhibits a slight deviation from the NYC demand curve, involving frequent fluctuation between high and low wind power output.
In this study, we developed a conceptual understanding of V2G but also developed and researched different computational tools for optimal simulation. Hence, we believe that any potential improvements in computational methods to calculate the feasibility of V2G should include predictive modeling techniques such as the Markov Decision Process (MDP) used in reinforcement learning to forecast the total number of EVs needed.

Nonetheless, we believe that, given the appropriate application, the potential of V2G can still play a significant role in the energy transition. Our analysis suggests that it may not be best suited as a primary storage solution for large-scale energy surplus but instead for managing local and dynamic demand response situations, such as unforeseen energy deficits. The actual value of V2G lies in its mobility and decentralized energy storage, not in maintaining a constant power balance. Future mobility studies should also consider other modes of electric transportation, like electric buses, due to their high battery capacity and predetermined schedules and routes. It would also be interesting to incorporate spatial mobility elements and contrast them

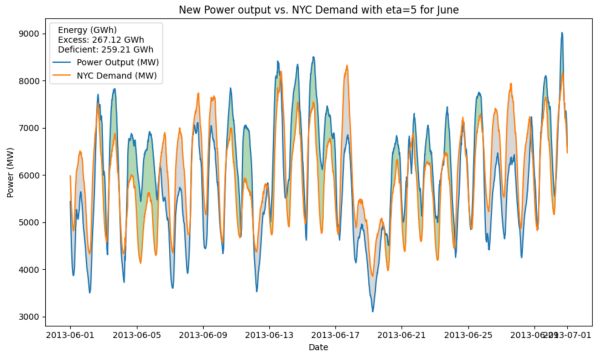
with local grid capabilities. Gaining knowledge about the locations of the majority of parked cars can as well provide valuable information regarding the potential impact on the power grid's sustainability. Based on our analysis, it is evident that a wind farm with a capacity of 9000MW or even 13000MW alone cannot meet the annual electricity demand of New York City, even without V2G technology. Thus, it is crucial to integrate other renewable energy sources, such as hydroelectric power and photovoltaics, particularly in the summer months, which could effectively decrease storage requirements.

Lastly, V2Gs proper integration is limited by its technical viabilities and the need for more financial incentives for electric vehicle (EV) owners to supply energy back to the urban grid. Most people purchase EVs for transportation purposes, and they would only be interested in providing power back to the grid if it is convenient and financially advantageous. Relying on individual participation to meet New York City's energy demand could pose significant risks unless the integration process considers monetary incentives and minimal technical hurdles to ensure ease of use for citizens. Therefore, it may be necessary to reconsider decentralized storage options at a more detailed level. For example, instead of focusing on cities and intra-city mobility, we could explore the potential of Vehicle-to-Home (V2H) programs or other grid-scale or user-level storage solutions.

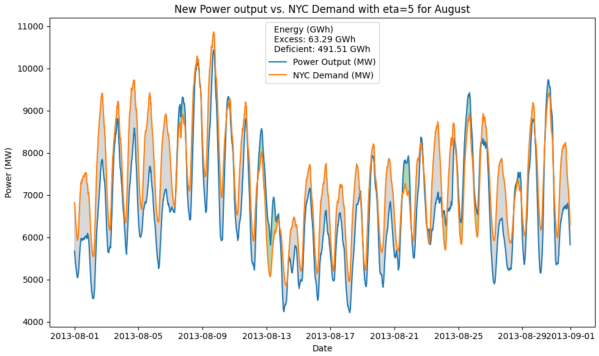
Appendices



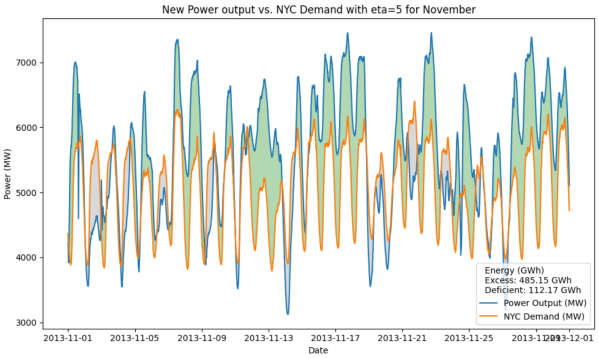
(a) Power output with $\eta = 5$ and C=13000MW in January



(b) Power output with $\eta = 5$ and C=13000MW in June



(c) Power output with $\eta = 5$ and C=13000MW in August



(d) Power output with $\eta = 5$ and C=13000MW in November

Figure 1: New Power output generation with $\eta = 5$ against NYC demand

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